

Where Confidence Gets Laundered

The Label Carries the Signal, This Reader Discards It

A thesis on state handoffs between frozen language models:
what crossed, what did not, and what an economy of state would need

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Abstract

When one language model hands work to another it writes a message, and the message usually carries a confidence word. Shi et al. [2026] call what happens next *confidence laundering*: the receiver treats a confident-looking guess as a fact. This thesis measures that claim with one addition the handoff literature had not made: it scores the message before it scores the reader. On 377 SQuAD items, between a frozen K2-Horizon-3.7B sender and a frozen 7B receiver, a confidence word scored directly as a label predicts the sender’s semantic entropy at AUROC .955 on clean passages, and the receiver that reads that word, a base model under one completion prompt, lands at .552. The signal was in the message; this reader threw it away. A mapped KV cache, which carries no label at all, leaves the receiver at .585 on the same items, nearly where reading the passage itself leaves it (.597), and moves the receiver’s confidence within an item because of the cache’s content (Spearman .280 against .004 for another item’s cache). The cache is a real channel: 63% on an 8-name binding test with chance at 12.5%, the wrong episode’s cache followed 64% of the time, and 53.0 SQuAD F1 against 54.7 for text. It is also 72 MiB per 512-token message. The second half of the document is a three-year program, one kill criterion per chapter. The reusable contribution is the audit: score the message, then the reader. Any paper that reports a verbal-confidence baseline should report the label-only AUROC next to the receiver AUROC.

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What this document claims

Three lists. A claim appears in exactly one of them. The tag that marks it here is the tag that marks it wherever it recurs in the text. SHOWN means a measurement with a control, reported in docs/paper/main.tex or in a json under analysis/cloud/. SUGGESTED means the direction is consistent across variants or seeds but at least one interval includes zero. BET means there is no evidence yet: a dated prediction from docs/agent-communication-futures-2026-09-21.md or goal/GOAL.md, or a plan in Part IV.

SHOWN (number, then control)

- **The label carries the signal; this reader discards it.** Confidence word scored as a label: AUROC .955 / .942 / 1.000 (clean / contradicted / removed) against the sender’s semantic entropy; the same word read by the receiver: .552 / .542 / .592. Control: the deranged cache at .536 / .488 / .564 and question-only at .532 / .479 / .603 bound what “no message” gives (Chapters 8 and 9).
- **Content crosses.** 8-name binding test, $n = 1000$: trained projector 63.0%, chance 12.5%. Controls: the wrong episode’s cache 4.6% with the wrong value followed 63.7% of the time; a projector trained with a deranged sender 23.4% (ridge base 21.7%); raw injection 14% (Chapter 4).
- **Content crosses on real passages.** SQuAD F1 53.0 for the mapped cache against 54.7 for text and 16.6 for the question alone. Controls: wrong passage’s cache 13.8, ridge 14.3, deranged-sender projector 15.1 (Chapter 6).
- **The receiver’s confidence moves because of the cache’s content.** Within-item Spearman between receiver entropy change and sender SE change (clean $\rightarrow$ removed): cache .280, text .322, word .146. Control: another item’s cache .004, which shares the question (Chapter 9).
- **Cache beats the word on contradicted passages.** Cache – word = [.008, .117], paired passage-grouped bootstrap; cache – question-only = [.065, .180] on the same variant (Chapter 8).
- **Text beats the cache on removed passages.** Cache – text = [−.091, −.002] (Chapter 8).
- **The recipe crosses one family boundary and fails on one pair.** Qwen3-4B $\rightarrow$ K2-7B: 71.7 $\pm$ 4.5% over three seeds; K2-0.9B $\rightarrow$ 7B: 17.0% (deranged 14.0%) (Chapter 5).
- **The sender’s state predicts its own failure.** Probe on the 3.7B’s hidden state: AUROC .749 on 600 SQuAD items against its own semantic entropy at .689; .876 [.813, .944] at layer 18 on 127 agent steps against trajectory metadata at .704 (Chapter 10, analysis/cloud/cascade/curves.json, nested_3.7b.json).

- **Retention cannot tell a channel from a prompt.** The deranged-sender projector retains 0.77 of the continuation-loss gap, 0.72 of it with its input swapped; the binding test puts it near chance (Chapter 4).
- **The message costs 144 KB per token,** 72 MiB per 512-token prefix, about 604 ms on a 1 Gbps link against 24 ms of receiver compute saved; int4 keeps 75.0% / 51.6 F1 at 36 KB per token; dropping layers collapses even under a filler null (Chapter 7).
- **On SQuAD no trigger makes a handoff pay:** the receiver reading the passage is 55.7 F1 against the sender's 52.2 (Chapter 10).

SUGGESTED (direction consistent, an interval includes zero)

- The ordering text $\gtrsim$ cache $>$ word holds on all three variants (.597 / .615 / .671 vs .585 / .605 / .624 vs .552 / .542 / .592), but cache $-$ word includes zero on clean ($[-.027, .094]$) and removed ($[-.024, .086]$), and cache $-$ text includes zero on clean and contradicted.
- The cache keeps more of the sender's confidence drop than the word (drop ratio .54 vs .30), a relative drop in $P(\text{gold})$ with no interval.
- The receiver recovers the sender's alternatives about as well from the cache as from the passage (belief-shape Spearman .199 vs .193 on clean passages; controls .052 / .042); no interval, and the arms are not separated from each other.
- Where the cache arm loses what it loses is not localised: sender state .739 / .706, one-position probe of the message .557 / .579, receiver via cache .643 / .563, via text .650 / .654.
- Two receiver-side training objectives did not move the loss (drop ratio .56 and .51 against .54), which points at the sender side, without yet testing it.

BET (no evidence yet; each carries a date or a kill criterion)

- The unit of computation is becoming a population of models, and its protocol will move from text to state (goal/GOAL.md, north star sentences 1–2; no date, re-worded only).
- A self-improving system will be a population handing state to each other and learning from what comes back (north star sentence 4, marked a bet there too).
- A sender-side adapter trained on returned state raises the sender more than returned text at equal tokens (Chapter 16; kill: $(b) - (a) \leq 0$ on three seeds).
- On a task with a sender–receiver gap large enough to pay for a handoff, state beats text at equal measured cost only intra-GPU (Chapter 17; price sheet declared before the run).
- A receiver-side detector tells a genuine mapped cache from a tampered one at AUROC $> .7$ (Chapter 15).
- From the futures document, three dated predictions this program can score: (11) a within-family KV projector loses more than 10 points of retention after a post-training update to the receiver, by 2027-12-31; (8) no replicated same-model multi-agent gain over 5 points at matched tokens with no private information, by 2028-12-31; (3) no production system passes latent state between two vendors across an organisational boundary, by 2030-12-31.
- Latent state stays inside a shared-weights boundary for legal rather than technical reasons; if so, the population this thesis studies is one owner's models (Chapter 3).

The handle. One pair of numbers runs through the document: **.955 against .552**. The first is a confidence word scored as a label on clean passages; the second is the same word scored through the receiver. Everything else is context for that pair.

Part I

The problem and the bet

The field holds a claim with two readings; this part says what measurement tells them apart.

1. A population of models, and the claim they launder confidence

1.1 The field’s standing belief, in its own words

Shi et al. [2026] titled their position paper *Confidence Laundering in Agent Systems: Why Uncertainty Needs a Latent Carrier*. The claim is in the title. When a small model hands its answer to a larger one as text, with a word such as “confidence: low” attached, the larger model reads the answer and not the word; the hedge is washed out and a guess arrives downstream looking like a fact. Their remedy is in the title too: carry uncertainty in a latent, not in a sentence. They supported it with one experiment, Qwen-1.5B in a HotpotQA tool loop, a semantic-entropy label from 20 samples, and four handoffs scored by how well a downstream component ranks the high-entropy cases: answer only, answer plus a score, answer plus a textual description of uncertainty, answer plus a latent carrier. The latent carrier won. Their motivating picture is an edge–cloud stack that escalates from a small on-device model to a large one.

1.2 The crack

A low receiver-side score through a text handoff has two causes, and no handoff study had separated them.

Either the message does not carry the sender’s uncertainty, or the reader does not use it. The first is a statement about a channel; the second is a statement about one model under one prompt. They call for different remedies, they generalise differently, and they cost one line of code to tell apart: score the message itself against the same target the receiver is scored against. If the message scores high and the receiver scores low, the loss is at the reader. If both score low, the loss is in the message. Shi et al. [2026] report the receiver’s number only, with no content-free control, so their result is consistent with both readings. So is every text-versus-latent comparison we found through 2026-09-23 (Appendix C).

This thesis is that one line, run properly: four kinds of handoff between the same two frozen models on the same items, with three controls per passage variant, and the message scored before the reader on every arm that has a message to score.

1.3 The two protocols side by side

Table 1.1 puts the position paper’s measurement next to ours. The target is the same, AUROC against the sender’s semantic entropy. The difference is in what sits around it.

The table is not a complaint about a position paper. It is the reason the thesis exists: the field’s standing belief was stated with a measurement that could not distinguish its two readings, and this document is the measurement that can.

Table 1.1: The measurement of Shi et al. [2026] and the measurement in this thesis. Their column is taken from the description in docs/paper/main.tex, Related Work, except the two cells marked †, which come from docs/lit-confidence-transfer-2026-09-22.md and carry that note’s “unverified” flag (not checked against the PDF); ours from main.tex, Sections 2–5.

	Confidence (2606.20662)	Laundrying	This thesis
target	sender semantic entropy, 20 samples, binarised†		sender semantic entropy, 10 samples, median per variant
sender → receiver	Qwen-1.5B; downstream probe at blocks 7 / 14 / 28		K2-Horizon-3.7B → K2-Horizon-7B, both frozen
handoffs	answer; +score; +textual uncertainty; +latent carrier		passage as text; answer + word; + number; + candidate list; mapped KV cache
items	one HotpotQA tool loop; N not stated†		377 SQuAD items × 3 passage variants (clean, contradicted, removed)
content-free controls	none		deranged cache, zero cache, moment-matched random cache, question only, per variant
intervals	none reported		paired, passage-grouped bootstrap on three cache differences
message scored directly	no		yes: label-only AUROC for the word and the number; linear probe for the cache
common-cause control	no		within-item intervention: question fixed, sender’s passage changed

1.4 What the measurement found, in one paragraph

On clean passages the confidence word, scored directly as a label, reaches AUROC .955 against the sender’s semantic entropy; the receiver that reads it reaches .552. **[SHOWN]** The signal was in the message. The reader, a base model under one completion prompt with no instruction to hedge, did not turn it into its own spread. That is what laundrying looks like from inside, and it is a property of a reader, not of text. A mapped KV cache with no label at all leaves the receiver at .585 on the same items, and when the sender’s passage is changed with the question held fixed the receiver’s entropy moves with the sender’s through the cache (Spearman .280) and not through another item’s cache (.004). **[SHOWN]** The cache carries something the receiver uses. What it carries is, by the parsimonious reading, the evidence; the receiver recomputes its own uncertainty from it, and no probe in this document isolates the sender’s spread crossing as such. **[SUGGESTED]** **The label carries the signal, the reader discards it; the cache carries the evidence, the reader recomputes.**

1.5 Contributions

Three findings, each with the chapter that defends it.

First, the receiver-level ordering text $\gtrsim$ cache $>$ word, measured for four handoffs on 377 items with content-free controls per variant and paired passage-grouped intervals on the three cache differences, with cache – word excluding zero on one variant of three and the within-item intervention (.280 / .322 / .146 / .004) ruling out the question as a common cause (Chapters 8 and 9).

Second, the **label-only audit** and the **two-level report** it belongs to: a message-level signal (the message scored against the sender target) next to a receiver-level signal (the receiver scored against the same target), with their difference the **read-out gap**. Applied here it relocates the text arm’s loss from channel capacity to read-out, .955 against .552, for this receiver (Chapter 9). This is the reusable artefact of the thesis.

Third, the instrument’s calibration: a channel whose content the receiver demonstrably reads (63.0% on an 8-name binding test, chance 12.5%, the wrong episode’s cache followed 63.7% of the

time), that recovers 53.0 SQuAD F1 against 54.7 for text, crosses one family boundary and fails on one pair, and costs 72 MiB per 512-token message (Chapters 4 to 7).

The names are used from here without re-explanation. *Message-level signal*, *receiver-level signal*, *read-out gap*, *label-only audit*. When “the audit” appears alone it means the label-only audit: score the confidence label directly against the sender target, then score the receiver that read it.

2. Text by inertia, and what state could carry

2.1 Why the unit of computation is becoming a population

A single model answering a single prompt is no longer the thing that gets deployed. Routers choose among models; cascades escalate from small to large; speculative decoding runs a draft model inside a target model’s loop; agents call other agents as tools. Every one of these is a population of models whose members hand work to each other, and in every one of them the handoff is text. [BET] The north star in `goal/GOAL.md` states this as its most certain sentence. Treated here as a bet: the evidence is a reading of the serving stack, not a measurement.

Text is the protocol by inertia. It is what the models were trained to read, what humans can inspect, what logs can keep. None of those is a reason it should be the protocol between two models that share nothing with a human reader. The question this thesis asks is narrower than “should the protocol be state”; it is “what does a state handoff carry that a text handoff does not, measured against controls, between two frozen models that we can afford to run.”

2.2 Three things text cannot carry

`goal/GOAL.md` names three: the computation already done, the shape of a belief (which alternatives, with what weight), and intermediate representations that are not words. Of the three, this thesis has measured the first and tried to measure the second.

The computation already done is what the binding test and the SQuAD run measure. When the sender has read a passage and the receiver reads only the question, the receiver answers at 53.0 F1 through the mapped cache against 16.6 from the question alone. [SHOWN] The sender’s reading crossed. On an A100 the receiver would have spent 48 ms re-reading 512 tokens; the projector takes 24 ms. [SHOWN] That is the saving, and Chapter 7 says what it costs in bytes.

The shape of a belief is what the uncertainty protocol tries to measure, and the result is that the receiver’s confidence tracks the sender’s through the cache about as well as through the passage itself, which is consistent with the evidence crossing and the receiver recomputing, and no experiment here isolates the shape crossing as such. [SUGGESTED] Chapter 9 reports the diagnostic closest to the shape, the correlation between the sender’s candidate frequencies and the receiver’s first-token mass on those candidates, and it separates the arms from the controls, not from each other.

Intermediate representations that are not words are not measured here at all. The setting is context reuse, the sender’s cache of a passage prefix, not a mid-task handoff of an agent’s reasoning or tool state, and nothing in this document says whether anything survives a sender that has reasoned at length [Liu et al., 2026]. [BET] Chapter 17 says what it would take to measure it.

2.3 The read-out gap, named, with a toy example

Fix a sender target u_S : a binary label, here “the sender’s semantic entropy is above its median.” Fix a message m that the sender produces. Score two things against u_S with the same statistic, AUROC.

1. The **message-level signal**: the AUROC of a fixed function of m alone. For a confidence word

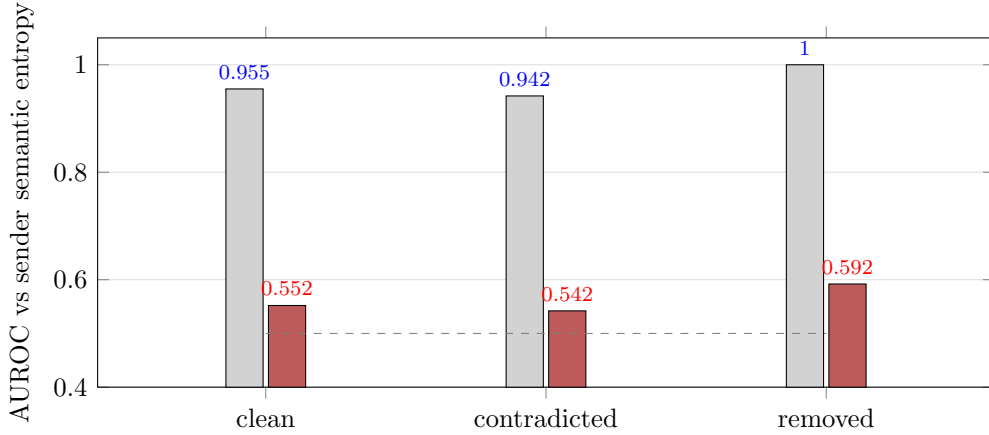
it is the rank of the word (low > medium > high). For a number it is the number. For a cache, which has no label, it is a linear probe on the mapped message, named as such wherever one appears.

2. The **receiver-level signal**: the AUROC of the receiver’s first-token entropy after it has read m .

Their difference is the **read-out gap**. It is a difference of two AUROCs, not an information decomposition; the cache’s two levels use different instruments, so the gap is comparable within an arm and not across arms.

The toy example needs no numbers beyond definitions. Take a sender whose label is a perfect function of u_S (the label says “low” exactly when $u_S = 1$). Its message-level AUROC is 1 by construction. Now take two readers. Reader A copies the label into its own entropy: high entropy when the label says low. Its receiver-level AUROC is also 1, and the read-out gap is 0. Reader B ignores the label and answers from its own knowledge of the question. Its entropy is independent of the label, its receiver-level AUROC is .5 (the definition of chance for the statistic), and the read-out gap is .5. Both readers received the same message. A study that reports only the receiver-level number would say that the text handoff “laundered” the sender’s uncertainty for reader B, and would be unable to say whether the message or the reader was at fault. The audit says which: the message scored 1.

The measured version is Figure 2.1. The label here is not perfect, because it is built from terciles over 587 items and scored against a median over the 377 kept items, but it is close: .955 / .942 / 1.000. The reader lands at .552 / .542 / .592. This reader is reader B.



■ confidence word scored as a label (message-level) ■ receiver reading the word (receiver-level)

Figure 2.1: The handle. The same confidence word scored two ways on 377 SQuAD items per variant: directly as a label against the sender’s semantic entropy, and through the receiver’s first-token entropy after it read the word. Dashed line: chance. Numbers from `docs/paper/main.tex`, Table 5; the 1.000 on removed passages is a degeneracy of the construction (22% of items tie at the threshold).

2.4 Belief, question, finding, mechanism, remedy

Belief. Text handoffs launder uncertainty; a latent carrier is needed.

Question. Message or reader?

Finding. .955 against .552 on clean passages; .942 against .542 under a contradiction; 1.000 against .592 with the answer removed. **[SHOWN]**

Mechanism. For the text arm, read-out. For the cache arm, the entry is “not localised”: linear probes at four points of the chain (sender state .739, mapped message at one position .557, receiver

via cache .643, via text .650 on clean passages) do not say where the cache arm loses what it loses, and the parsimonious reading is that evidence crosses and the receiver recomputes. [SUGGESTED]

Remedy. One line. Score the message, then score the reader. Report both. The smallest remedy, as when Darcet et al. [2024] fixed attention artefacts by adding a few spare tokens rather than a new model: no new model here either, no new objective, one extra number in every table that has a verbal-confidence baseline.

3. What an economy of state would need

3.1 Cost, value, fraud

Once state can be handed off there is something to trade, and a traded good needs three things: a price, a way to be defrauded, and an institution that makes trade safe when the buyer cannot read the good. `goal/GOAL.md` puts it as cost, value, fraud $\rightarrow$ price, reputation, audit. This chapter says which of those the repository has measured, with the tag each deserves, and keeps the economy itself where it belongs: in the BET column.

Price. A price needs bytes, latency and accuracy gained. All three are measured for one pair. The mapped cache is $36 \text{ layers} \times 8 \text{ KV heads} \times 2 \times 128 \times \text{bf16} = 144 \text{ KB}$ per token, 72 MiB for a 512-token prefix, against about 1 KB of text. [SHOWN] The projector takes 24 ms where the receiver's re-read takes 48 ms; over a 1 Gbps link the message itself takes about 604 ms (int4: about 151 ms). [SHOWN] Accuracy gained is 53.0 F1 against 54.7 for text on SQuAD and 39.9 ± 2.1 against 45.8 on partitioned HotpotQA. [SHOWN] So the price sheet already exists for this pair, and what it says is that state clears above text only when bytes are free, that is, intra-GPU. Chapter 17 declares that expectation before any run that could change it.

Fraud. Confidence laundering is the fraud: a confident-looking state that is wrong. The repository has measured the buyer's exposure. Handed the wrong episode's cache, the receiver answers with the wrong episode's value 63.7% of the time; handed the output of a projector trained with a deranged sender, it answers at 23.4%, above chance. [SHOWN] The receiver believes what it is handed. Nothing yet detects a tampered cache; Chapter 15 is the chapter that would.

Audit. The audit primitive is the thing this thesis contributes: score the message, then the reader. It is an audit of a text good today. For a state good the message-level instrument is a linear probe, and the probe bounds what one position carries (.557 / .579), not what the map preserves. [SUGGESTED] An audit a buyer could run on a cache without the seller's cooperation is a bet (Chapter 15).

Reputation. Not measured. A sender whose reliability a buyer can track over time requires a sender whose reliability changes, and the three projector checkpoints in the repository are static. [BET] The futures document's prediction 11 (a within-family projector loses more than 10 points of retention after a post-training update to the receiver, by 2027-12-31) is the cheapest way to make a sender's reliability move, and Chapter 17 folds reputation and drift into one experiment for that reason.

3.2 The counter-bet: the wire splits at the weights

The repository holds a second document that argues against the north star. `docs/agent-communication-futures-2026-09-21.md` predicts that dense latent channels will run inside a shared-weights boundary and that text, signed and rewritten into a standard form, will cross every

organisational boundary, less because of engineering than because of law: recordkeeping, liability, privacy and antitrust rules all assume a message someone can read. Its fifteen dated predictions mature between 2027 and 2030. Three of them bear directly on this program and are listed in the BET box at the front: prediction 11 on drift, prediction 8 on multi-agent gains without private information, prediction 3 on cross-vendor latent state in production.

If the predictions hold, the population studied here is one owner’s models: a defensible thesis, and not the agent economy of the north star’s third sentence. The disagreement is not resolved here. It is scheduled: Chapter 17 runs the drift test that prediction 11 names, and Chapter 18 records the date on which each prediction is scored.

3.3 What this document does not claim

Nothing here says a KV cache is the right protocol for a population of models; at 72 MiB per message it is a measurement instrument, not a deployment, and the one experiment that would make it worth its bytes, a task where the sender’s state and not its passage is the point, has not been exhibited. Nothing here says the sender’s uncertainty crosses the channel as such; the within-item test licenses “the receiver’s confidence moves because of the cache’s content” and no more.

Part II

What we established

Five chapters, one instrument: a cache the receiver demonstrably reads, priced in bytes, then the uncertainty it carries.

4. Content transfer and its controls

The figure shows one thing: the trained projector sits far above chance, the wrong episode’s cache sits below it, and a projector trained with a deranged sender sits near it.

[SHOWN]

4.1 Why the instrument comes first

Before the receiver’s confidence can track the sender’s, the cache must carry the passage. Every later chapter stands on this one, and so does the audit: a message-level signal for the cache is only meaningful if the cache is a channel. The audits of Cheng et al. [2026] and Zhang and Emu [2026] set the bar. The first finds that on natural benchmarks most of a relayed cache’s gain is receiver-side computation (MedQA: +0.4 of +14.7 points is content-specific), and that content transfer appears only when the receiver needs sender-private information. The second decomposes a latent channel’s effect into “sender content” and “any message present.” The binding test, the derange arm and the deranged-sender projector are built for those two objections.

4.2 The binding test

Each episode assigns one of eight single-token values to eight nonce names (“Noma is amber. Vela is indigo. ...”). Only the sender reads the facts. The receiver is asked “What colour is Vela?” and the eight values are scored. On the 8-name test (Table 4.1) the trained projector reaches 63.0% against chance 12.5 and the wrong episode’s cache 4.6. [SHOWN] Handed the wrong episode’s cache the receiver answers with the wrong episode’s value 63.7% of the time, and its accuracy falls below

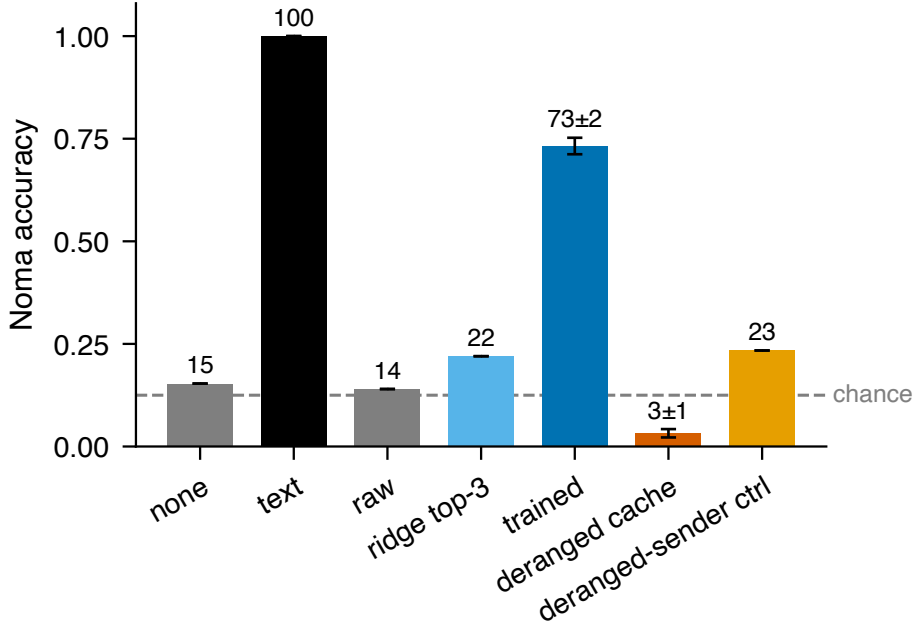


Figure 4.1: Noma accuracy by arm on the 6-name variant ($n = 300$). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ($n = 1000$, 8-name variant). Dashed line: chance (12.5%).

Table 4.1: Binding transfer, 8 names $\times$ 8 values, $n = 1000$ episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

projector	none	text	raw	project	derange	follow	content
ridge top-3 (closed form)	10.6	100	14	21.7	11.4	18.2	10.3
+ residual, 1500 steps	10.6	100	14	63.0	4.6	63.7	58.4
residual, deranged sender	10.6	100	14	23.4	11.3	22.7	12.1

chance: it is not ignoring the cache, it is reading it. Matching shape alone (*raw*) does nothing (14%). The easier 6-name variant ($n = 300$), used for the seeds, the ablations and Figure 4.1, gives 71.3 / 73.0 / 75.3% over three training seeds (mean 73.2, sd 2.0). [SHOWN] The same protocol with cities and animals (8 names, $n = 500$) gives 61.8% and 67.2%, and padding the facts with up to 512 filler tokens does not reduce accuracy (Chapter 5).

Two ablations on the 6-name variant say where the recipe’s accuracy comes from (Appendix A): freezing the ridge and training only the residual gives 59.0%, so letting the linear map move during training is worth about 14 points; replacing the correlation-selected top-3 source layers with the aligned-layer map gives 32.7%, so which sender layers feed each receiver layer is the largest design choice. [SHOWN]

4.3 The deranged-sender projector

A projector trained with the identical recipe but with the sender reading text unrelated to the receiver’s continuation cannot learn a content channel, only a soft prompt. It reaches 23.4% on the 8-name binding test (Table 4.1, row 3). It is not content-free: its ridge base is weakly content-dependent (ridge alone 21.7%), so it controls for the *learned* part only. [SHOWN] This is the “any message present” term of Zhang and Emu [2026], isolated for a trained map.

4.4 Where the binding lives

Table 4.2: Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere. $n = 500$, 8-name variant: the jsons (analysis/cloud/noma_layers*mlp.json) record 39 fact tokens, the layout of Table 4.1’s run, where docs/paper/main.tex labels this table 6-name; the jsons win. The “all” column is the $n = 1000$ row of Table 4.1, not an $n = 500$ run. Accuracy in percent.

layers	0–5	0–11	0–17	12–23	24–35	30–35	18–35	all
accuracy	7.2	8.0	10.4	28.6	30.8	7.6	51.8	63.0

The first half of the network carries nothing usable on its own, the second half alone carries most of it, and no six-layer band suffices (Table 4.2, plotted in Figure 5.2). [SHOWN] The binding is read out from a distributed set of later layers. The filler used to complete the cache in these runs is the deranged episode’s cache, a plausible but wrong state; Chapter 7 repeats the pruning under a neutral filler and gets the same collapse.

4.5 Retention cannot tell a channel from a prompt

Table 4.3: Continuation retention is the wrong headline. $\text{Retention} = (\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

projector	retention	retention, sender text deranged
ridge top-3	−0.30	−0.66
+ residual	1.23	0.31
residual, deranged sender	0.77	0.72

Following the closed-form-map literature [Heo et al., 2026] the paper also reports retention (Table 4.3; Figure 4.2). Theirs is an accuracy ratio; ours is the fraction of the continuation-loss gap recovered on 128 fresh held-out sequences deduplicated against training text (on 256 such sequences: 1.19 for the trained projector, −0.38 for the ridge), so the table critiques our own metric, not theirs. The deranged-sender projector recovers 77% of the oracle gap, and almost all of it (0.72) survives when its input is deranged at test time. The trained projector’s 1.23 decomposes into about 0.3 of the same prompt-like effect and about 0.9 that disappears when the sender’s text is swapped. [SHOWN] Retention above 1 is not evidence of communication, as Cheng et al. [2026] objected; the binding test with its derange arm is. The finding is a positive artefact: a metric that cannot tell a channel from a prompt, and a test that can.

With that established the instrument is calibrated: the cache carries the passage, and any uncertainty signal measured through it is measured through a channel that could in principle carry one.

5. Across families, and where the recipe fails

The figure shows the recipe working on three pairs, one of them across a family boundary, and not on the fourth.

[SHOWN]

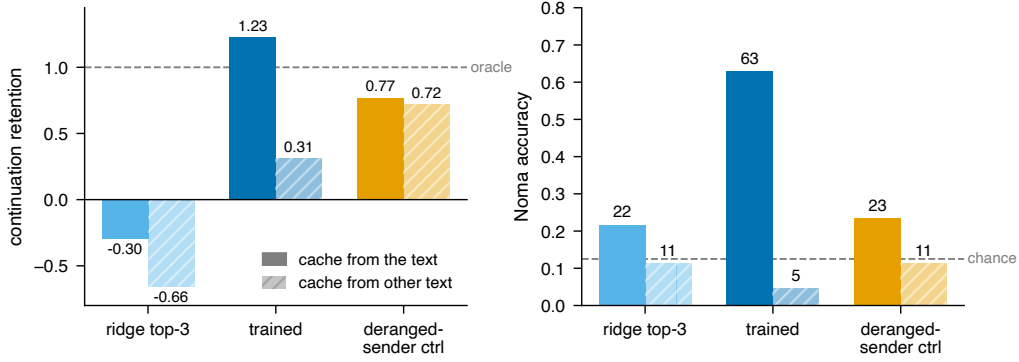


Figure 4.2: Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ($n = 1000$, 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the deranged-sender projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

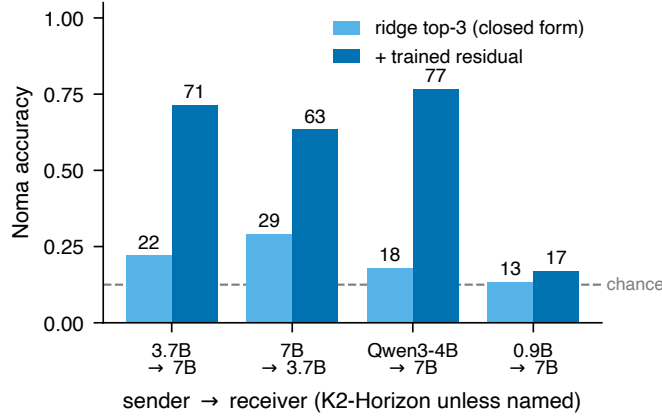


Figure 5.1: Noma accuracy (*project* arm) for the closed-form ridge map and the trained projector on four sender–receiver pairs ($n = 300$, 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

5.1 Other attributes and reversed roles

The channel is not one attribute and not one-way. With cities and animals in place of colours the trained projector gives 61.8% and 67.2% with the deranged-sender control at 15.4% and 12.4% (Table 5.1). With the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate: the same pattern at slightly lower accuracy. [SHOWN] The reverse map matters later: it is the return path that Chapter 16 would train on.

5.2 Four pairs, one recipe

With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128, matched KV geometry, but a different vocabulary and pre-training) receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge 18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7 / 70.3 / 68.0% over three training seeds (mean 71.7, sd 4.5; deranged $\leq 4.7\%$, follow rate 80% for seed 0), SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for the receiver reading the text. [SHOWN] Cross-family transfer is as strong as within-family on the binding test and within 7 F1 on passages, and the linear map alone does nothing across families.

Table 5.1: Binding transfer with city and animal attributes ($n = 500$, 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names, $n = 300$).

setting	projector	none	project	derange	follow
city	ridge top-3	14.2	19.8	11.8	18.4
city	trained	14.2	61.8	6.4	60.2
city	deranged-sender ctrl	14.2	15.4	11.4	18.6
animal	ridge top-3	12.2	12.8	12.0	11.2
animal	trained	12.2	67.2	5.2	64.6
animal	deranged-sender ctrl	12.2	12.4	12.8	9.8
reversed	ridge top-3	11.7	29.0	12.3	31.3
reversed	trained	11.7	63.3	5.7	63.0

Table 5.2: Four sender–receiver pairs, same recipe. Noma 6-name variant, $n = 300$, chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ($n = 300$; text ceiling 54.7). K2-Horizon unless named. Qwen3-4B row: proj. and F1 are mean $\pm$ sd over three training seeds (76.7 / 70.3 / 68.0; F1 48.0 / 51.9 / 44.9); der. is the worst seed, follow is seed 0. The raw arm is undefined for K2-0.9B (head dimension 64).

sender $\rightarrow$ receiver	raw	ridge top-3			+ trained residual			F1
		proj.	der.	follow	proj.	der.	follow	
3.7B $\rightarrow$ 7B	14.0	22.0	10.7	30.3	71.3	4.3	71.7	53.0
7B $\rightarrow$ 3.7B	11.7	29.0	12.3	31.3	63.3	5.7	63.0	—
Qwen3-4B $\rightarrow$ 7B	15.7	18.0	13.0	17.0	71.7 $\pm$ 4.5	≤ 4.7	80.3	48.3 $\pm$ 3.5
0.9B $\rightarrow$ 7B	n/a	13.3	13.0	10.7	17.0	14.0	11.3	24.2

5.3 Where it fails

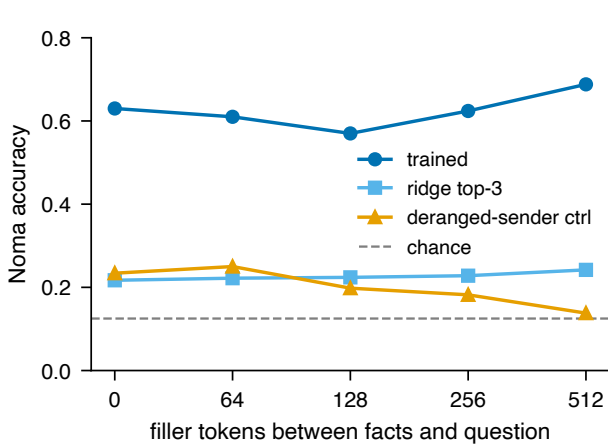
For K2-Horizon-0.9B, which differs from the receiver in vocabulary, layer count and head dimension (64) at once, the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1. **[SHOWN]** This parameterisation fails on that pair. Which of the three differences matters is untested, and cross-head maps exist in other work, so the failure is of this recipe, not of the pair; Chapter 10 returns to it as a finding, and Chapter 13 says what a second data point would cost.

5.4 Distance

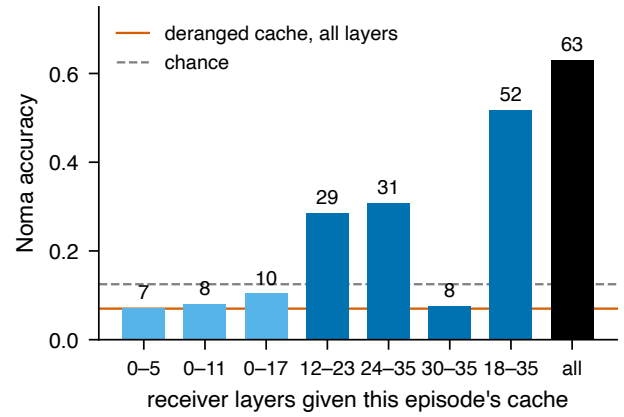
Table 5.3: Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant; $n = 1000$ at pad 0, $n = 500$ otherwise.

projector	0	64	128	256	512
ridge top-3	21.7	22.2	22.4	22.8	24.2
trained	63.0	61.0	57.0	62.4	68.8
deranged-sender ctrl	23.4	25.0	19.8	18.2	13.8

Padding the facts with neutral filler so that the question is 64 / 128 / 256 / 512 tokens after them gives 61.0 / 57.0 / 62.4 / 68.8% (Table 5.3, Figure 5.2). No decay; the best value is at the training prefix length of 512. **[SHOWN]**



(a) Accuracy against filler tokens between the facts and the question (8-name variant; $n = 1000$ at 0, $n = 500$ otherwise).



(b) Accuracy when only the labelled band of receiver layers receives this episode's mapped cache (the deranged episode's cache fills the rest; Table 4.2). Solid line: the deranged cache in all layers; dashed: chance.

Figure 5.2: Distance sweep and layer-band injection.

Matched geometry crosses a family boundary; mismatched geometry does not cross within one. The recipe has a domain, and the domain is the KV shape, not the name on the checkpoint. [SHOWN]

6. Real passages and different contexts

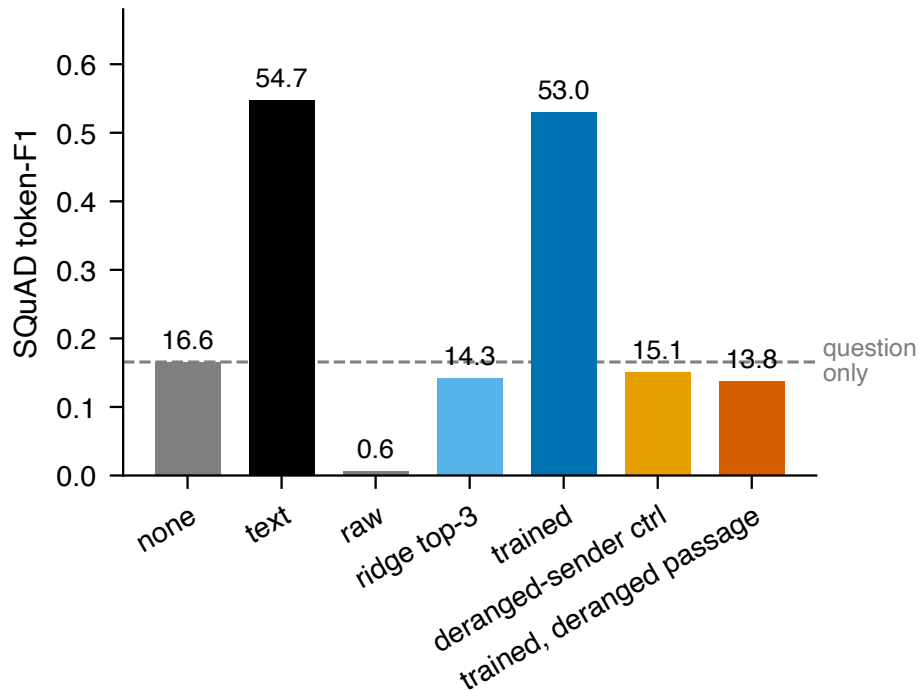


Figure 6.1: SQuAD token-F1 by arm ($n = 300$). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage's cache all sit at or below question-only.

The figure shows the mapped cache within two F1 of the text ceiling and every control at or below the question-only floor.

[SHOWN]

6.1 SQuAD

Table 6.1: Real passages: SQuAD v1.1 validation, $n = 300$ (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

receiver sees	none	text	raw	ridge	trained	deranged- sender ctrl	trained, derange arm
F1	16.6	54.7	0.6	14.3	53.0 (51.8)	15.1	13.8
gold log-prob / token	-3.57	-0.90	-11.2	-3.88	-1.74	-3.77	-4.18

The same projectors, never trained on questions, applied to SQuAD v1.1 validation: the sender reads the passage (≤ 512 tokens), the receiver reads only the question and answers greedily (≤ 8 tokens, stopped at newline), token-F1 against the gold spans. On F1 the trained projector recovers 96% of the gap between question-only and reading the passage (53.0 at $n = 300$; 51.8 at $n = 600$); on gold log-probability it recovers 69% (Table 6.1). **[SHOWN]** The wrong passage’s cache (13.8), the ridge map (14.3) and the deranged-sender projector (15.1) all sit at or below question-only (16.6). On passages, as on the binding test, the channel carries content, not a bias toward answering. The gap between the F1 recovery and the log-probability recovery is the first sign of something Chapter 8 measures directly: the cache path answers nearly as well as the text path and is less confident while doing it.

6.2 Different contexts: HotpotQA

Table 6.2: HotpotQA bridge questions, $n = 400$. Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1; the mapped cache is mean $\pm$ sd over three training seeds (37.9 / 39.7 / 42.1), the deranged cache their range.

receiver additionally gets	F1
nothing (own paragraph + question)	30.1
the sender’s paragraph as text	45.8
the sender’s paragraph as a mapped cache	39.9 $\pm$ 2.1
a deranged mapped cache	29.9–32.1
the sender’s paragraph through the ridge map	26.3
a zero cache	20.9

This is the setting of Rossi et al. [2026] and the one setting in the repository where the sender holds evidence the receiver lacks, which Cheng et al. [2026] identify as the only condition under which relayed caches carry content on natural tasks. The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9 / 39.7 / 42.1 over the three training seeds (mean 39.9, sd 2.1), 63% of the gap, while the deranged cache gives 29.9–32.1 (no help), the ridge map 26.3 and a zero cache 20.9 (Table 6.2). **[SHOWN]** The channel carries the *other* paragraph, and only the trained map does. The cache costs 5.9 F1 against text here, the number Chapter 17’s price sheet has to beat.

7. What the message costs

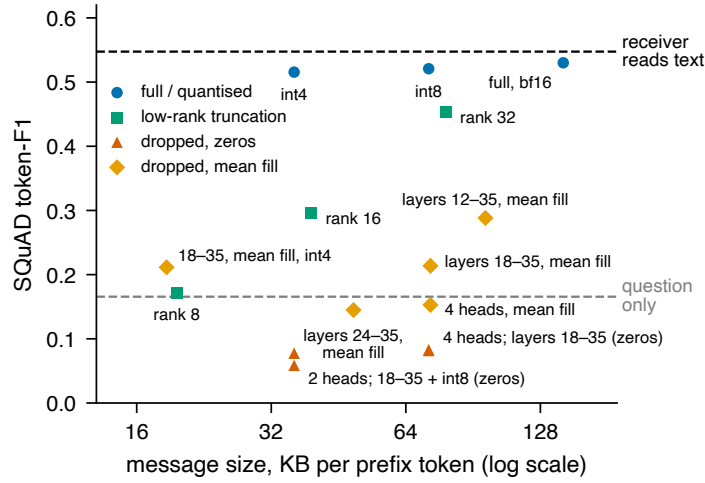


Figure 7.1: Message size against SQuAD F1 for every compression of the mapped cache in the zero- and mean-fill sweep ($n = 300$). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

The figure shows one free move (quantisation) and no other: every way of cutting the message below a quarter of its size cuts the content with it.

[SHOWN]

7.1 Bytes and time

Table 7.1: What the message costs. Bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant, $n = 300$, seed-2 projector. “filler” fills the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length. The full sweep, including zero and mean-vector fills and smaller projectors, is in Table 7.2.

message	KB / token	Noma	SQuAD F1
full mapped cache, bf16	144	75.3	53.0
int4 fake-quantised	36	75.0	51.6
layers 18–35 only, filler elsewhere	72	28.7	16.1
4 of 8 heads, filler elsewhere	72	26.7	12.8

The full mapped cache is $36 \text{ layers} \times 8 \text{ heads} \times 2 \times 128 \times \text{bf16} = 144 \text{ KB per token}$, 72 MiB for a 512-token prefix, against about 1 KB of text; the projector has 151 M parameters. [SHOWN] Int4 fake-quantisation keeps binding accuracy (75.0 vs 75.3%) and SQuAD F1 (51.6 vs 53.0) at a quarter of the bytes, but this is fake quantisation, not a packed transport (Table 7.1). Nothing else is free. Dropping layers or heads collapses transfer even when the untransmitted layers hold the receiver’s own cache of a neutral text, a valid episode-free state: layers 18–35 alone give 28.7% on the 6-name variant and 16.1 F1, 4 of 8 heads 26.7% and 12.8 F1. Zero and mean-vector fills collapse the same way (Table 7.2), so the loss is missing content, not broken attention, and with this projector the message cannot be cut by layers. [SHOWN] Low-rank truncation keeps the 6-name binding test (rank 32 / 16: 75.0%) but not passages (F1 45.3 / 29.6).

Latency on one A100: the receiver re-reading 512 tokens takes 48 ms, the projector 24 ms. But at 72 MiB per message a 1 Gbps link needs about 604 ms (int4: about 151 ms) against 24 ms of receiver

compute saved, so the compute advantage does not carry over to a network handoff at this size. [SHOWN] A reader who can get the sender’s uncertainty by re-reading 1 KB of text should; the channel is worth its bytes only where the sender’s state, not its passage, is the point, and the paper has not exhibited such a case. That sentence is the deployment limit, and Chapter 17 is built around it.

7.2 The full sweep

Table 7.2: Every compression of the seed-2 projector’s mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant, $n = 300$, percent; SQuAD: $n = 300$, token-F1. “filler” rows fill the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length.

spec	KB / token	Noma project	Noma derange	SQuAD F1
none (full, bf16)	144	75.3	2.3	53.0
int8	72	76.0	2.7	52.1
int4	36	75.0	3.7	51.6
rank 32	78.7	75.0	2.3	45.3
rank 16	39.3	75.0	2.3	29.6
rank 8	19.7	51.7	7.0	17.2
layers 18–35, zeros	72	24.3	9.7	8.3
layers 18–35, zeros, int8	36	24.7	9.7	7.8
4 heads, zeros	72	25.7	8.7	8.2
2 heads, zeros	36	22.3	9.7	5.9
layers 12–35, mean fill	96.4	35.0	11.0	28.8
layers 18–35, mean fill	72.7	33.7	11.7	21.4
layers 18–35, mean fill, int4	18.7	33.3	11.7	21.1
layers 24–35, mean fill	48.9	29.3	12.0	14.5
4 heads, mean fill	72.7	23.0	11.3	15.3
layers 12–35, filler	96	30.7	11.7	18.9
layers 18–35, filler	72	28.7	12.0	16.1
layers 18–35, filler, int4	18	30.0	11.7	16.9
layers 24–35, filler	48	25.3	10.0	13.3
4 heads, filler	72	26.7	10.0	12.8
projector bottleneck 32 (70 M params)	144	64.3	5.7	53.7
projector bottleneck 16 (62 M params)	144	61.3	7.7	51.6

A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. The filler rows are the structural null: the receiver’s own cache of a neutral text is a valid, episode-free state, so a collapse under it is missing content rather than broken attention. [SHOWN] Three nulls (zeros, mean fill, filler) agree, which is why Chapter 10 files layer pruning under findings rather than under things still worth trying with this projector.

8. Uncertainty: how much survives each handoff

The figure shows the read-out gap as a vertical distance: the hollow markers for the word and the number sit far above their bars, and the cache, which has no marker, sits level with the text. [SHOWN]

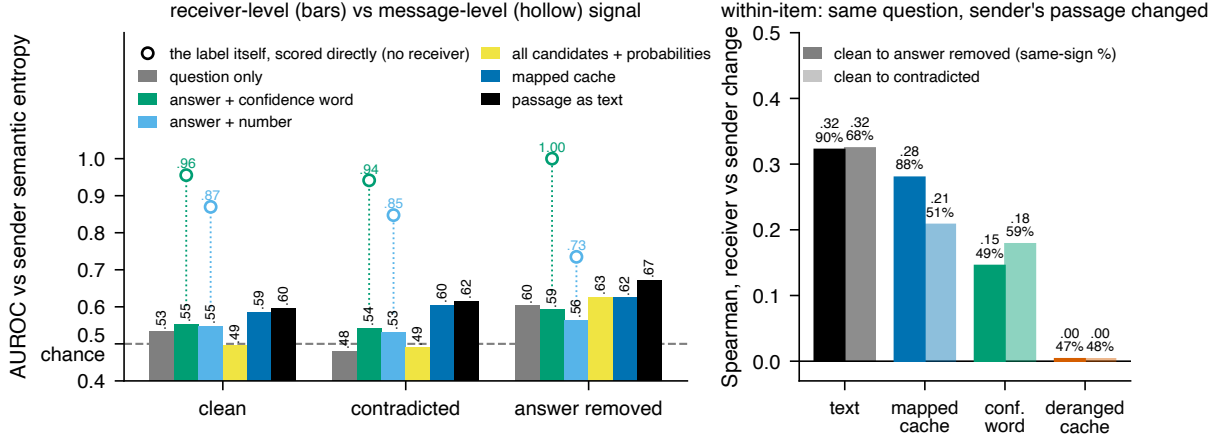


Figure 8.1: Left: receiver-level signal (bars: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, 377 items) for six handoffs per passage variant, with the message-level signal of the label itself drawn as a hollow marker above the word and number arms; the vertical distance from marker to bar is the read-out gap. The cache has no label and no marker. Right: within-item intervention, the question held fixed and the sender’s passage changed; bars are the Spearman correlation between the receiver’s entropy change and the sender’s semantic-entropy change, annotated with the same-sign rate, which is bounded by the 91.5% of items on which the sender’s entropy rises.

8.1 The protocol

Following the knowledge-conflict and semantic-entropy protocols [Longpre et al., 2021, Xie et al., 2024, Farquhar et al., 2024], each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold sentence deleted). The 377 items (344 distinct passages) on which the receiver reading the clean passage reaches token-F1 $\geq .5$ are kept. The sender’s uncertainty is its semantic entropy (SE) over 10 sampled answers (mean 0.98 / 1.18 / 1.96 over the 377 kept items on the three variants, so the sender does notice); the target is SE thresholded at its median per variant. Ten samples make SE coarse: 38 / 40 / 28 distinct values per variant, and on removed passages 22% of items sit exactly at the threshold. The receiver’s uncertainty is its entropy at the first answer token.

Four messages. Two are functions of the passage alone, $M = f(c)$, with the question unseen by the sender: *text* and *mapped cache*. Two require the sender to answer the question, $M = f(c, q)$: *verbal* (the sender’s greedy answer and “confidence: high / medium / low”) and its variants with a number (the sender’s sample agreement, e.g. “0.8”) or the full candidate list (every answer cluster with its sampled frequency). The confidence word is an oracle label built from the sender’s *actual* SE terciles, favourable to the text channel; the sender does not verbalise its own confidence. The receiver’s prompt is a plain completion with no chat template, no few-shot examples and no instruction to hedge; it was not ablated, and Chapter 10 says what that leaves open. Content-free controls per variant: a deranged cache (another item’s), a zero cache, a moment-matched random cache, and question only. Appendix B has the definitions in full.

8.2 The table

8.3 Every arm as two rows

A reader should leave with the split, not the aggregate. Table 8.2 rewrites Table 8.1 that way: a label row, a reader row, and the read-out gap, per arm, with the row in which the loss sits named in prose.

Table 8.1: Sender uncertainty through each handoff; 377 SQuAD items on which the receiver reading the clean passage reaches token-F1 $\geq .5$. AUROC is of the receiver’s first-token entropy against the sender’s semantic entropy (thresholded at its median per variant); the drop ratio is the receiver’s $P(\text{gold})$ loss from clean to removed relative to the text path’s. The two “scored directly” rows are the message-level signal: the label itself ranked against the same target, with no receiver. Two projectors trained with uncertainty objectives are in Table 10.1.

receiver sees	$P(\text{gold})$			entropy		AUROC vs sender SE			drop ratio
	clean	contra.	removed	clean	removed	clean	contra.	removed	
text	.437	.290	.162	1.14	3.03	.597	.615	.671	1 (def.)
text + answer + confidence word (verbal)	.253	.193	.171	2.88	2.80	.552	.542	.592	.30
text + answer + numeric confidence (sample agreement, e.g. “0.8”)	.258	.197	.170	2.93	2.93	.548	.532	.563	.32
text + all candidate answers with their sampled probabilities	.264	.210	.144	2.68	2.63	.495	.489	.626	.44
mapped cache	.279	.249	.130	1.78	3.16	.585	.605	.624	.54
deranged cache	.105	.106	.105	3.88	3.88	.536	.488	.564	0
zero cache	.034	.033	.037	6.41	6.18	.496	.481	.502	0
moment-matched random cache	.005	.004	.005	7.24	7.30	.457	.472	.580	0
question only	.127			3.69		.532	.479	.603	0
<i>message-level signal (no receiver)</i>									
confidence word, scored directly as a label	—			—		.955	.942	1.000	—
numeric confidence, scored directly	—			—		.870	.848	.735	—

Table 8.2: Each handoff as a two-level report, clean / contradicted / removed. The message-level row for the cache is the one-position linear probe of Chapter 9, run on the 587-item set rather than the 377 (clean / removed only), so its gap is not comparable across arms and is written in grey. Numbers from Table 8.1 and Chapter 9.

arm	level	clean	contradicted	removed
confidence word	message (label)	.955	.942	1.000
	receiver	.552	.542	.592
	read-out gap	.403	.400	.408
numeric confidence	message (label)	.870	.848	.735
	receiver	.548	.532	.563
	read-out gap	.322	.316	.172
mapped cache	message (one-position probe, 587 items)	.557	—	.579
	receiver (377 items)	.585	.605	.624
	read-out gap	not localised; see Chapter 9		
text (passage)	message	no label to score		
	receiver	.597	.615	.671

For the word and the number the loss sits in the reader row: the label is near-perfect by construction and the receiver reading it is within .07 of question-only on every variant (.020 / .063 / .011). **[SHOWN]** For the cache the loss cannot be assigned to a row: the receiver recovers more (.643, by the probe protocol of Chapter 9) than a one-position probe of its message carries (.557), so the message carries more than one position reads, and the entry is “not localised.” **[SUGGESTED]** For text there is no message-level row at all, which is the point: the passage is not a summary of the sender’s belief but the evidence it was formed from, and the receiver forms its own.

8.4 Result

On clean and contradicted passages the ordering of receiver-level signal is text, then the mapped cache, then every summary of the sender’s belief; on removed passages the candidate list matches the cache (.626 vs .624) and question-only (.603) beats the word (.592). The cache beats the confidence word by +.03 to +.06 AUROC, and its drop ratio is .54, that is $(.279 - .130) / (.437 - .162)$, where the word’s is .30. Writing the confidence as a number does not help the text channel (.548 / .532 /

.563; drop ratio .32), and neither does the strongest text message that can be built from the sender’s samples, every answer cluster with its frequency (.495 / .489 / .626; drop ratio .44). [SHOWN]

Intervals were computed for three differences, each a paired bootstrap over passages (344 passages, 2,000 resamples). Cache – word excludes zero on contradicted passages ([.008, .117]) and not on clean ([−.027, .094]) or removed ([−.024, .086]); cache – question-only excludes zero on contradicted ([.065, .180]) and is borderline on clean ([−.005, .111]); cache – text includes zero on clean ([−.066, .047]) and contradicted ([−.064, .047]) and excludes it on removed ([−.091, −.002]), where the text path is above the cache. The direction is consistent; the size is modest and established on one variant of three. [SUGGESTED]

8.5 Controls, per variant

The content-free controls do not sit at one level, and an earlier draft that said they did was corrected by review (Appendix C). On clean and contradicted passages they lie at .46–.54 (question only .532 / .479; deranged .536 / .488; random .457 / .472; zero .496 / .481) against the cache’s .585 / .605. On removed passages question-only alone gives .603, the random cache .580 and the deranged cache .564 against the cache’s .624: the question already carries most of the signal when the answer is gone, and question-only there beats the word (.592). The cache’s increment over question-only is therefore +.053 on clean, +.126 on contradicted and +.021 on removed passages, and only the middle one is established. [SHOWN]

8.6 Behaviour under contradiction and removal

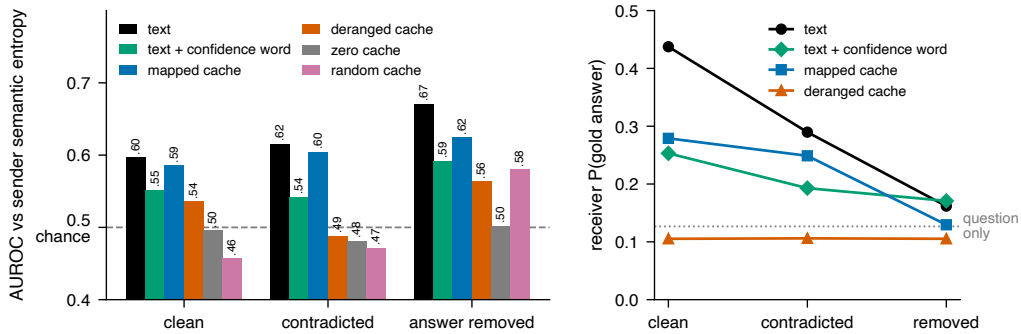


Figure 8.2: Left: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, per handoff and passage variant (377 items), including the zero and moment-matched random caches. Right: the receiver’s $P(\text{gold})$ from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.

When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise. Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ($P(\text{counter}) .32 > P(\text{gold}) .29$), while the cache path keeps the original ahead ($.25 > .22$); across all conditions the cache path is less confident than the text path on clean passages ($P(\text{gold}) .28$ vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache (Figure 8.2). [SUGGESTED] What this chapter cannot say is why the ordering is what it is; that is the next chapter’s job.

9. Where the loss occurs

Three predictions, each written here as a prediction followed by its test, and three agreements: the word arm’s loss is at the reader; the cache arm’s is not localised. Two of the three tests carry a control or an interval; the third is a direction with no interval. Analysis is kept apart from results here so the count of evidence is visible. The tests were not pre-registered: the label-only audit and the within-item intervention were computed on 2026-09-25 after the fifth and sixth reviews asked for them (Appendix C), and the predictions were written down after the numbers existed.

9.1 Three tests that the reader discards the label

Prediction 1. If the loss on the word arm is in the message, the word scored directly against the sender target will be low, near the receiver’s .552. If the loss is in the reader, the word will score high and the receiver low.

Test 1: the label-only audit. The confidence word is built from the sender’s own SE terciles, so scored as a label it predicts the binary target almost perfectly without any receiver: AUROC .955 / .942 / 1.000 on the three variants, the numeric label .870 / .848 / .735 (Table 8.1, bottom). [SHOWN] These numbers are a property of the construction, a tercile split over all 587 items against a median split over the 377 kept items, and the 1.000 on removed passages is a degeneracy: 22% of items tie at the threshold and the tercile boundary coincides with it. The audit is a sanity check, not a finding; the finding is the other bar: the receiver’s first-token entropy after reading that word reaches .552 / .542 / .592. The text channel is not capacity-limited here.

Prediction 2. If the receiver’s confidence moves because of what the sender saw, and not because hard questions make both models uncertain, then changing only the sender’s passage (question fixed) will move the receiver’s entropy with the sender’s through a channel that carries the passage, and will not move it through a cache that shares the question but not the passage.

Test 2: the within-item intervention. The variants allow a paired test with the question held fixed and only the sender’s passage changed (clean → removed; the sender’s SE rises by 0.98 on average). Per item, the Spearman correlation between the receiver’s entropy change and the sender’s SE change is .280 through the mapped cache, .322 when the receiver reads the text, .146 through the confidence word, and .004 through a cache from another item, which shares the question and rules out the common cause (Figure 8.1, right). [SHOWN] The same-sign rates (88 / 90 / 49 / 47%) are weaker evidence than they look: the sender’s SE rises on 91.5% of items, so any arm whose entropy simply rises on removal scores near that ceiling and an arm with no mean shift sits at a coin flip; same-sign measures the mean shift, the Spearman the per-item covariation. Under a contradiction the effect is smaller (cache .208, text .325, word .179, deranged .004). The receiver’s confidence moves with what the sender saw, and it moves because of the cache’s content. It licenses no more than that: the text path tracks at least as well, so the parsimonious reading is that the evidence crosses and the receiver recomputes its own uncertainty from it.

Prediction 3. If the reader discards the label, then on passages where the answer is gone, a receiver with no message at all should do no worse than a receiver handed the word.

Test 3: question-only against the word on removed passages. Question-only reaches .603, the word .592 (Table 8.1). [SUGGESTED] No interval was computed for this pair; it is a direction, .011 wide, and it is counted as such. The label, which scores 1.000 directly on this variant, adds nothing the question did not already give the receiver.

Three tests in agreement, two of them with a control or an interval behind them. The read-out failure is a property of the label channel for this reader, not of handoffs in general. It is also why the capacity argument of the paper’s earlier draft (Chapter 10) cannot be the mechanism of the verbal arm’s shortfall: a channel that delivers a .955 label is not the bottleneck.

9.2 Shape of the belief

Sender SE and receiver first-token entropy are different measurements; a diagnostic closer to the belief itself is the *shape*: per item with at least two sampled answer clusters, the Spearman correlation between the sender’s cluster frequencies and the receiver’s first-token mass on those candidates. On clean passages ($n = 323$) it is .199 through the mapped cache, .193 from the text, .166 from the explicit candidate list, and .052 / .042 for the deranged cache and question only: there the receiver recovers the sender’s alternatives about as well from the cache as from the passage, and no better from a list of them. Under a contradiction ($n = 351$) the order reverses (cache .153, list .201, text .234), and on removed passages every arm is near zero. [SUGGESTED] No interval was computed for these correlations, and they are small enough that the diagnostic separates arms from the controls, not from each other.

9.3 Four-point probes: the cache arm’s loss is not localised

The same standardised logistic probe ($C = 0.05$, five passage-grouped folds, no inner selection; the semantic-entropy-probe protocol of Kossen et al., 2024) for the binary sender-SE target at four points of the chain, on inputs of 2560 / 2048 / 6144 / 4096 dimensions and on the 587 items (the full set before the F1 filter, so these AUROCs are not directly comparable with Table 8.1). Clean / removed: the sender’s own last-prompt-token state, averaged over layers 12 / 18 / 24, gives .739 / .706; the mapped message at its last prefix position, averaged over layers and heads, .557 / .579 (with the question-only receiver state concatenated, .563 / .603); the receiver’s state after reading through the mapped cache .643 / .563; after reading the text .650 / .654; the question alone .530 / .591. [SHOWN]

Table 9.1: Four-point probes against the binary sender-SE target, 587 items, clean / removed. AUROC of a standardised logistic probe with passage-grouped folds. From docs/paper/main.tex, Section 5(d); analysis/cloud/night_a/diag_unc_clean.json and diag_unc_removed.json.

probed state	clean	removed
sender, last prompt token, layers 12 / 18 / 24	.739	.706
mapped message, last prefix position	.557	.579
mapped message + question-only receiver state	.563	.603
receiver after reading the mapped cache	.643	.563
receiver after reading the text	.650	.654
receiver, question only	.530	.591

Three things follow (Table 9.1). The sender holds more about its own uncertainty than any receiver state recovers. On clean passages the receiver reaches .643 through the cache, above the .557 that the probed message carries, so the message carries more than a one-position probe reads, and the cache leaves the receiver with as much as the text does (.643 vs .650); on removed passages the receiver (.563) falls below the probed message with the question (.603) and below the text (.654). These probes therefore do not localise the loss. [SUGGESTED] Two caveats bound them further. The message-level probe sees one position of 512, so .557 / .579 bounds what one position carries, not what the map preserves; a pooled read-out of the whole prefix is the obvious next probe and is Chapter 14’s first experiment, with a one-sided kill fixed in advance. And the sender probe saw the question as well as the passage, while the mapped message is passage-only, so sender-minus-message mixes question conditioning with map loss.

9.4 What the analysis licenses

For the text arm: read-out. For the cache arm: evidence crosses and the receiver recomputes, with the sender’s spread crossing as such neither shown nor excluded. The thesis’s title claim is therefore

about the label, and the cache is in the title only as the contrast that shows the receiver can be moved by a message when the message is the evidence rather than a summary of it. Nothing here says a text handoff cannot carry uncertainty; it says this base receiver, under one completion prompt, does not read the label it was handed.

Part III

What failed and what it taught

Six results that went against the design. Each closed a door a later chapter would otherwise have had to open.

10. Six negatives, filed as findings

None of the six is an apology; each is tagged with what it established.

10.1 The 0.9B pair: a recipe has a domain

K2-Horizon-0.9B differs from the 7B receiver in vocabulary, layer count and head dimension (64 against 128) at once. The same recipe that gives 71.3% on the 3.7B and $71.7 \pm 4.5\%$ on Qwen3-4B gives 17.0% here (deranged 14.0%, follow 11.3%, content transfer 3 points) and 24.2 F1 on SQuAD (Table 5.2). **[SHOWN]** What it taught: matched KV geometry, not family membership, is what this parameterisation needs. Qwen3-4B crosses a family boundary with matched geometry; a same-family sender with mismatched geometry does not cross. Which of the three differences matters is untested, and the repository has two data points, which is why the program treats “content transfer is a property of the triple” as a hypothesis with a kill (Chapter 13) rather than a conclusion.

10.2 Pruning under three null states: the message cannot be cut by layers

The layer-band table (Table 4.2) showed the second half of the network alone carries most of the binding; the natural next move was to send only those layers. Under three fills for the untransmitted layers (zeros, the per-layer mean vector, and the receiver’s own cache of a neutral text) layers 18–35 give 24.3 / 33.7 / 28.7% on the 6-name variant and 8.3 / 21.4 / 16.1 F1 (Table 7.2). **[SHOWN]** The fourth review had pointed out that a zero fill dilutes the softmax, so a collapse under zeros could be broken attention rather than missing content; the filler fill is a valid episode-free state, and the collapse survives it. What it taught: with this projector every receiver layer wants its own cache, and the 72 MiB figure cannot be halved by dropping layers. The by-layer route to a cheaper message is closed for this recipe; training-time layer dropout is the untested alternative.

10.3 Low rank: the binding test is too easy to price a message by

Low-rank truncation of each (positions $\times$ head-dim) matrix at rank 32 and 16 keeps the 6-name binding test at 75.0% and loses passages (F1 45.3 and 29.6; rank 8: 51.7% and 17.2 F1). **[SHOWN]** What it taught: a compression that passes the binding test can still destroy the passage, so the binding test alone cannot be used to price a message. Every compression row in Table 7.2 carries both columns for that reason, and Chapter 13’s sender catalogue requires both.

10.4 Two training objectives that did not move the loss

Table 10.1: Two projectors trained with an uncertainty objective (weight 1, 1,000 steps), scored exactly as in Table 8.1.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
mapped cache (reference, Table 8.1)	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
mapped cache, KL-trained projector	.272 / .241 / .118	2.38 / 3.76	.590 / .589 / .611	.56
mapped cache, sender-entropy-matched projector	.247 / .222 / .107	2.23 / 3.64	.596 / .587 / .627	.51

If the cache arm’s shortfall were an artefact of the projector’s training target, changing the target should move it. Two receiver-side objectives did not (Table 10.1). Adding $\text{KL}(\text{text-path} \parallel \text{cache-path})$ on the receiver’s next-token distributions (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1. Matching the receiver’s per-token entropy through the cache to the sender’s own per-token entropy on the training text holds binding accuracy (72.3%), drops SQuAD F1 to 45.3, and leaves the uncertainty measures where they were (.596 / .587 / .627; drop ratio .51). **[SHOWN]** What it taught: neither target contains the sender’s spread. The receiver’s own text path does not contain it either, and entropy matching is not belief matching. The untested fix is on the other side of the map: a separate reader $r(m, q)$ trained on the sender’s candidate distribution and kept apart from the receiver’s own decision. That is Chapter 14’s pooled reader.

10.5 The handoff policy on SQuAD: the trigger works, the task does not pay

Table 10.2: Handoff policy on 600 SQuAD items. Top: AUROC of each trigger at predicting the sender’s own failures ($F1 < 0.5$). Bottom: F1 of each arm when every item is handed off (rate 1). `analysis/cloud/cascade/curves.json`.

sender (3.7B) alone: F1 52.2, wrong on 35.5% of items	
trigger	AUROC
probe on the sender’s hidden state	.749
semantic entropy (10 samples)	.689
sample disagreement	.683
max-probability	.582
entropy	.584
receiver (7B) sees	F1 at rate 1
the passage as text	55.7
the mapped cache	51.8
the sender’s answer + confidence word	47.0

On 600 SQuAD items the 3.7B alone reaches F1 52.2 and gets 35.5% of items wrong. A logistic probe on its pre-action hidden state (last prompt token, layers 12 / 18 / 24; five-fold, not grouped by passage, $C = 0.1$) predicts those failures with AUROC .749, better than its own semantic entropy (.689), sample disagreement (.683), max-probability (.582) or entropy (.584) (Table 10.2). **[SHOWN]** The same kind of probe on 127 agent steps from 26 runs, nested leave-one-run-out, reaches .876 [.813, .944] at layer 18 against trajectory metadata at .704, and in the shortcut audit the probe alone gives .881,

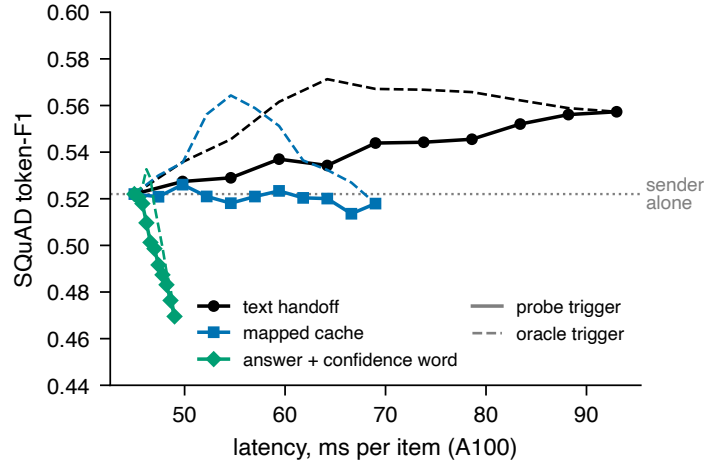


Figure 10.1: F1 against latency per item as the handoff rate goes from 0 (sender alone, 45 ms) to 1, for the three handoff arms under the probe trigger (solid) and an oracle trigger that hands off exactly the sender’s failures (dashed). 600 SQuAD items. Only the oracle gains; the probe-triggered text handoff climbs to the receiver’s own 55.7, the cache handoff stays flat.

metadata .703, both together .835 (`analysis/cloud/nested_3.7b.json`, `shortcut_k2-3.7b.json`). **[SHOWN]** The sender’s state knows when the sender will fail.

But the 7B reading the passage reaches only 55.7, the mapped cache 51.8 and the verbal handoff 47.0, so no trigger makes a handoff pay at equal latency (Figure 10.1): the probe-triggered text handoff climbs from 52.2 to 55.7 as the handoff rate goes to 1, the cache handoff stays flat at 52, and only an oracle trigger gains (56.4 at rate .4). **[SHOWN]** What it taught: the receiver is not enough better than the sender on this task for a handoff of any kind to be worth it, which is a fact about SQuAD and not about the trigger. The policy result waits for a task with a real gap, and Chapter 17 refuses to run until that gap has been measured.

10.6 The retracted log 3 deduction

Version 4 of the paper carried a theory box that deduced the text arm’s low receiver-level AUROC from a capacity bound: a message that is the argmax of the sender’s belief plus a k -level label takes at most $|Y|k$ values and so carries at most $\log |Y| + \log k$ nats about the belief. The bound is correct and limits information about the *distribution*; it does not limit a detector’s AUROC against a thresholded target, and a three-level label reaches .955 against the median split. The step from the bound to the AUROC was invalid, it was never the argument of Shi et al. [2026], and it was withdrawn on 2026-09-25 after two external reviews said the same thing (Appendix C). What it taught: the measurement does the work, and theory enters this document only as a toy that makes a prediction (Section 2.3), never as a proof that the measurement had to come out that way. What survives of the box is one modest statement: if the receiver’s recovered belief is within total variation ε of the sender’s and one-step rewards lie in $[0, R_{\max}]$, acting on the recovered belief loses at most $2R_{\max}\varepsilon$ in expected reward. That is what a decision test should measure, and no experiment here measures regret.

10.7 What would change the claim

Each limitation below is paired with the cheapest experiment that would close it, and the last one is the experiment that would overturn the main claim.

- *One projector seed through the confidence protocol.* The three seeds trained for the binding test have not been run through it. Cheapest fix: run them; `docs/design-confidence-2026-09-`

22.md prices the protocol at one A100 session, under \$15. If the seed sd of cache — word exceeds the .03–.06 effect, the cache-versus-word ordering drops from SUGGESTED to unshown.

- *One dataset, one sender–receiver pair, one uncertainty measure.* Cheapest fix: the 377-item protocol on the HotpotQA partitioned items, eval-only on cached states.
- *The four-point probes read one position.* Cheapest fix: a pooled read-out over all 512 prefix positions, with a one-sided kill fixed in advance (Chapter 14).
- *The deployment limit.* 144 KB per token; about 604 ms per message on a 1 Gbps link against 24 ms saved. No experiment closes this; the price sheet of Chapter 17 declares where state can clear and where it cannot.
- *The overturning experiment.* “The receiver does not read the label” is a property of one base model under one completion prompt: no prompt ablation, no few-shot examples, no chat template, no instruction-tuned receiver, and the candidate-list arm scoring below question-only on clean passages (.495 vs .532) says the text baselines are not the strongest a competent engineer could build. A receiver whose entropy follows “confidence: low”, an instruction-tuned or few-shot reader, would erase the contrast. That is the first experiment the program runs (Chapter 14), because it is the only one that can change the abstract.

Part IV

The program

Everything from here is a bet, and every bet has its kill written before its run.

11. The thesis sentence, the price sheet, and the order of the kills

11.1 The sentence, written as a conditional

A small frozen model that predicts its own failure from its hidden state, hands its KV state rather than text to a stronger model, and updates a sender-side adapter from what comes back, improves more per unit of compute than the same model with a text handoff or no handoff, and the gain is attributable to content that crossed, not to extra computation in either model, on the price sheet declared below and on a task whose sender–receiver gap pays for the handoff. [BET]

Everything after “improves” is a bet. The repository has built two of the loop’s three parts: a probe that tells a small model it is about to fail, and a channel whose content the receiver demonstrably reads. The third part, a learning rule on the sender side driven by what the receiver returns, is untouched, and it is the part that turns a handoff into a population that improves. The program makes that loop the object, measured with the controls the first paper built (deranged cache, deranged-sender projector, content-free fills, within-item intervention) on one A100 at a time.

11.2 The price sheet, declared before any run

Three link rates, fixed now, so that no later result can choose the rate that flatters it. [BET]

The expected outcome, stated from the shown numbers: state clears above text only intra-GPU, and only on time. The HotpotQA partitioned run then tests whether even that holds when the cache costs 5.9 F1 against text. If it does not, the economy chapter is about the instrument and the audit.

Table 11.1: Pre-registered price sheet. Every entry is a SHOWN number from Chapter 7 or Chapter 6 arranged into a plan; the “expected” column is the author’s bet from those numbers, written before Chapter 17 runs. The link time of 1 KB of text is not measured and is left symbolic. A fourth, intermediate link rate is to be computed from the same 72 MiB before the run and is left without a number here.

rate	bytes cost	time, state vs text (link + compute)	accuracy, state vs text	expected
intra-GPU	free	0 + 24 vs 0 + 48 ms	SQuAD 53.0 vs 54.7; HotpotQA 39.9 vs 45.8	state clears on time, loses on F1
1 Gbps, bf16	72 MiB vs 1 KB	604 + 24 vs link(1 KB) + 48 ms	same	text clears
1 Gbps, int4	36 KB per token	151 + 24 vs link(1 KB) + 48 ms	51.6 vs 54.7 (SQuAD)	text clears

11.3 The order of the kills

The cheapest experiments that can change the most are run first, and two of them are run before any new chapter is opened. [BET]

1. *Week 1, no GPU.* Price Chapter 17 from the SHOWN numbers alone across a grid of byte and latency prices and publish the one figure that shows where state beats text. This decides “intra-GPU only” before any job runs.
2. *Block 1 on GPU.* An instruction-tuned or few-shot receiver, plus a chat template and an explicit hedge instruction, on the existing 377 items for the word arm (Section 10.7). It is the only experiment that can change the abstract of the paper going to arXiv, and the only one that decides whether Chapters 14, 16 and 17 are about uncertainty or only about latency.
3. *Block 1, same session.* The three existing projector seeds through the confidence protocol, so the .03–.06 effect gets a seed-variance bar before anything is built on it.
4. *Block 2.* The drift test from Chapter 17 pulled forward: LoRA-update the 7B on a small held-out corpus and re-score the frozen seed-2 projector on the 6-name binding test and SQuAD without refitting. This is prediction 11 of the futures document, dated 2027-12-31; its cost is quoted in Chapter 18 in the design documents’ units.
5. *Block 3.* The tamper detector of Chapter 15 on the fraud set the repository already holds.
6. Only then the gap task, the return path, and the loop.

11.4 Pre-registration and the statistical floor

Cheng et al. [2026] pre-registered their audit and used TOST equivalence margins; docs/design-confidence-2026-09-22.md already fixed a ± 0.03 AUROC margin for “cache $\approx$ text.” Every chapter below states its kill before its run, pairs at the item level, reports paired passage-grouped bootstrap intervals, and treats “within the margin” as a result rather than a failure. [BET]

12. Program chapter 1. Knowing you cannot: the pre-action probe

Hypothesis. A linear probe on the sender’s hidden state before it acts predicts its own failure [BET] better than any statistic of its output and better than trajectory metadata; the signal survives a change of dataset; and it separates “the answer is not here” from “I cannot find it.”

Done. [SHOWN] SQuAD, 600 items: probe AUROC .749 against semantic entropy .689, sample disagreement .683, max-probability .582 (Table 10.2). Agent traces, 127 steps from 26 runs, nested

leave-one-run-out: probe .876 [.813, .944] at layer 18, metadata .704; shortcut audit: probe alone .881, metadata .703, both .835 (`analysis/cloud/nested_3.7b.json`, `shortcut_k2-3.7b.json`).

Next. [BET] Train on SQuAD clean, test on HotpotQA partitioned and on the agent steps without refitting; split the failure label into answer-removed against answer-present-but-wrong (the two kinds of “don’t know”, information against capability; Appendix C) and report per-split AUROC with passage-grouped folds. Eval-only on cached states.

Kill. Off-distribution AUROC falls to the output-statistics level (at or below .69 on the same items), or the two failure kinds are not separable (per-split AUROC within each other’s interval). Either way the trigger is a within-dataset shortcut, and Chapters 16 and 17 fall back to semantic entropy as the trigger. Orgad et al. [2025] report that error probes do not transfer across datasets, so the null here is a live one.

Status. Partly done. Both SHOWN numbers exist; the cross-dataset and two-kinds split do not.

13. Program chapter 2. The channel, and a priced catalogue of channels

Hypothesis. A learned per-head map from the sender’s KV cache to the receiver’s lets the receiver answer from what only the sender read; the gain is content, not a soft prompt or extra receiver computation; and content transfer is a property of the (sender, receiver, map) triple that KV-geometry and vocabulary features predict. [BET]

Done. [SHOWN] The binding test, the deranged-sender projector, SQuAD, HotpotQA partitioned, the reverse direction, the Qwen3-4B pair, the 0.9B failure, the retention critique and the cost sweep, all in Part II. This chapter has already passed its kill: milestone 1 of `docs/goal-2026-09-21.md` asked whether the deranged-sender projector’s retention reaches 80% of the trained projector’s (`goal/GOAL.md` records only “milestone 1 done”), and it is 0.77 against 1.23 (63%), with the binding test putting the control near chance.

Next. [BET] A **priced sender catalogue**. Every external channel cited in Appendix C (C2C, StateBridge, Interlat, the universal context-reuse layer, the closed-form ridge map) gets the same four columns the repository already measures for its own projector: bytes per token, latency against re-fill, binding-test accuracy with the derange arm, and audit AUROC on the fraud set of Chapter 15. The lab never trains a stranger’s projector: the submission artefact is a mapped-cache file for a fixed item set (the 300 Noma 6-name episodes, the 377 SQuAD confidence items, the 400 HotpotQA bridge items) plus the submitter’s own deranged-sender control, and the lab runs only the receiver side, which the design documents price at one A100 session per triple. Third-party maps enter only where their pair’s tokeniser alignment is already solved: the closed-form ridge of Heo et al. [2026] on a K2 pair first, then StateBridge.

Kill. For the catalogue’s own hypothesis: fit the same recipe on eight or more triples (the repository has two points: matched geometry crosses families at 71.7 ± 4.5 ; head dimension 64 fails at 17.0) and predict binding accuracy and F1 from geometry and vocabulary features. If those features do not predict transfer above a shuffled baseline, “content transfer is a property of the triple” is unsupported and the table is reported without the claim. For the channel as a deployment: if nothing below about 1 MB per message keeps SQuAD F1 within 5 of text, the channel stays a measurement instrument (Chapter 7 already closes layers, heads and rank for this projector).

Status. Channel done; catalogue not started. Per-pair engineering hours are budgeted next to GPU hours in Chapter 18.

14. Program chapter 3. Does the sender’s uncertainty cross, or is it recomputed?

Hypothesis. The sender’s belief shape (its candidate distribution) is recoverable from the mapped [BET] message by a reader trained for it, at a level above what the receiver recomputes from content alone and approaching what the sender’s own state carries.

Done. [SHOWN] Everything in Chapters 8 and 9: the four handoffs, the label-only audit, the within-item intervention, the belief shape, the four-point probes, and the two objectives that did not move the loss.

Next. [BET] Three experiments the paper names as next, in the order of Chapter 11: (i) the instruction-tuned or few-shot receiver for the word arm, the one result that would overturn the paper’s main claim; (ii) the three existing projector seeds through the confidence protocol, the cheapest missing control; (iii) a pooled reader over all 512 prefix positions, trained on the sender’s candidate distribution and kept separate from the receiver’s decision. Then the 377-item protocol on partitioned HotpotQA, the one setting where the sender holds evidence the receiver lacks, with one-step regret under the total-variation bound of Chapter 10 as a second score next to AUROC.

Kill. One-sided and fixed now: the pooled probe must exceed .643 on clean passages (the receiver’s own recovery through the cache) for the “loss is not in the reader” reading of the cache arm to stand, and must exceed the one-position .557 / .579 by more than its passage-grouped interval to count at all. If the instruction-tuned receiver reads the word at or above the cache’s AUROC, or if any reader’s receiver-level AUROC comes within .1 of the .955 label signal, the claim shrinks to “content crosses and the receiver recomputes,” Chapter 16 drops the uncertainty term, and the probe remains the only carrier of “I cannot.” If the seed sd of cache – word exceeds the .03–.06 effect, the ordering is reported as unshown.

Status. Partly done; (i)–(iii) not run.

15. Program chapter 4. Fraud and audit: a confident wrong state, and a buyer who can tell

This is the one chapter whose value survives state losing to text on accuracy. “Authenticate the cache as you would any other input,” the paper’s own impact statement, is the audit primitive of the north star’s cost / value / fraud triple.

Hypothesis. A receiver can tell a sender’s genuine mapped cache from a deranged or norm-matched [BET] tampered one from the cache alone, better than a text-side audit of the same number at matched cost, and the detector holds against a sender trained to look confident.

Done. [SHOWN] The receiver believes what it is handed: follow rate 63.7% on the wrong episode’s cache, and the deranged-sender projector answers at 23.4%, so a confident wrong state is accepted today (Table 4.1).

Next. [BET] A receiver-side detector (a linear probe on the injected cache and on the receiver’s early-layer state) trained on {mapped, deranged, moment-matched random, zero, norm-matched tampered}, scored as AUROC with passage-grouped folds and as TPR at a fixed FPR. Two additions make it a claim about state rather than about probes. The *same-source baseline* (Appendix C): the sender’s own failure-probe score (AUROC .749 against the sender’s own failure, Table 10.2, already measured) sent as a text field against a probe on the mapped cache; the audit claim is only about state if the cache-side detector beats the text-side audit of the same number at matched cost. The *strategic sender*: one projector fine-tuned with a reward for looking confident (low receiver entropy) regardless of evidence, with the norm-matched attack recorded in the futures document as the reference; the detector is tested against it, which also gives Chapter 17’s reputation a non-static sender to learn, which the three current checkpoints are not.

Kill. Detector AUROC at or below .7 on the tampered class: then the channel cannot cross a trust boundary, the futures document’s reading holds (latent stays inside one firm’s weights), and the thesis is bounded to a population under one owner. If the cache-side detector does not beat the text-side audit of the same number, the chapter reports the detector as a receiver-side safety result and drops the word “state” from its title.

Status. Not started; the fraud set exists.

16. Program chapter 5. The return path: learning from what comes back

This is the spine of the second paper and the chapter where the literature has nothing. Distillation work observes that student maps contract predictive variance [Cui and Pei, 2026]; cascades threshold a scalar and re-printfill; nobody trains the sender on what a receiver gives back in state.

Hypothesis. A sender-side adapter trained on the receiver’s returned state (its answer plus its cache mapped back through the reverse channel) raises the sender’s standalone accuracy on held-out items more than distilling the receiver’s text answers at equal tokens, and more than self-training on the sender’s own answers. [BET]

Bar set by. [SHOWN] The reverse map already works (7B $\rightarrow$ 3.7B, 63.3% on the 6-name binding test, Table 5.1), and on partitioned HotpotQA the receiver holds what the sender lacks (45.8 against 30.1 as text, Table 6.2).

Next. [BET] Three arms, receiver frozen, sender given a small adapter. (a) Text return: sender fine-tuned on the receiver’s answers. (b) State return: sender fine-tuned with the receiver’s cache mapped back as a prefix, answer as target. (c) Self-return control: sender fine-tuned on its own answers, the “any message present” term of Zhang and Emu [2026] transposed to learning. Measure sender standalone F1 and the Chapter 12 probe AUROC before and after, three seeds, passage-grouped splits, paired bootstrap intervals; SQuAD first because the items and caches exist, HotpotQA partitioned second. Smallest version: 600 items, a fixed number of adapter steps per arm, all arms and seeds in one quarter’s budget (Chapter 18).

Kill. (b) $-$ (a) ≤ 0 within the paired interval on all three seeds, or (b) and (c) gain alike. Then “learning from what comes back” is text distillation under another name, and the thesis keeps the probe and the channel but drops the population claim to “a cascade with a better trigger.” The title loses the word “improve.”

Status. Not started.

17. Program chapter 6. Closing the loop on a task with a real gap, while the good decays

Reputation is only meaningful under drift, and drift is the one mechanism in reach that makes a sender’s reliability change, so they are one experiment.

Hypothesis. On a task where the receiver beats the sender by a margin large enough to pay for **[BET]** the handoff, the probe-triggered state handoff dominates text handoff at equal measured cost on the intra-GPU row of Table 11.1; after rounds of Chapter 16’s rule the handoff rate falls while accuracy holds; and when the receiver is updated mid-stream, a running estimate of the sender’s reliability beats a fixed price.

Bar set by. **[SHOWN]** The negative: on SQuAD the 7B reading the passage is only 3.5 F1 above the 3.7B (55.7 against 52.2), so no trigger pays; the probe-triggered text handoff climbs to 55.7 at rate 1, the cache handoff stays flat near 52, only the oracle gains (56.4 at rate .4). Measured costs: sender read 45 ms, projector 24 ms, re-prefill 48 ms; 72 MiB over 1 Gbps about 604 ms (int4 about 151 ms) against 24 ms saved.

Next. **[BET]** Step 1, before any training: measure the sender–receiver gap on three candidate tasks (HotpotQA partitioned; a harder QA set; a partially observable act / observe / ask / stop text task, ALFWorld-style with no robot (Appendix C), where the receiver must pick act / observe-more / ask / stop so that the price of a message includes the cost of a wrong action and the value of a kept alternative, not only F1) and keep the first with a gap of at least 15 F1 or its equivalent in regret. Step 2: accuracy-against-cost curves with cost in measured latency and in bytes (int4 packed, not fake-quantised), arms = state / text / verbal, triggers = probe / semantic entropy / random / oracle; `cascade.py` and `policy.py` reused unchanged. Step 3: run Chapter 16’s rule for k rounds and replot. Step 4, the decay: LoRA-update the receiver on a small held-out corpus mid-stream, re-score the frozen projector without refitting (the drift test, prediction 11), and give the buyer a stream in which the sender’s reliability actually changes; compare a running calibration against a fixed price. Where the task is sequential, evaluation is by branched rollouts with same-model control forks, the requirement of Gonuguntla [2026], and the environment is built only after Chapters 14 and 16 have returned at least one positive price-sheet result.

Kill. No candidate task in reach at this scale shows a gap (then the thesis is about the instrument, not the economy); or state never beats text at equal cost on any handoff rate (the kill written in `docs/design-cascade-2026-09-24.md`); or the handoff rate does not fall after learning. Any one of these removes the word “improve” from the thesis title. The drift step is labelled as a test of predictions 8 and 11 that the author expects to lose, so a negative result is a scored prediction rather than a dead chapter.

Status. Not started; step 1 is the first thing after the kills of Chapter 11.

18. Horizon, what kills the whole program, and what it costs

18.1 Six, twelve, thirty-six months

[BET] All three horizons are plans.

To 2027-04. arXiv v1 of the paper out (due 2026-10-01 by `goal/GOAL.md`) and the paper submitted to ICML 2027 (submission 2027-01-28, the quarter rung in `goal/GOAL.md`) or COLM 2027 (`goal/LOG.md`'s first choice; the two files disagree and neither has been updated). Chapter 14 closed either way: the instruction-tuned receiver for the word arm, the three seeds through the confidence protocol, the pooled reader. Chapter 12's cross-dataset and two-kinds split done on cached states. The drift test (prediction 11) run once. Chapter 15's detector on the existing fraud set with the same-source baseline. Chapter 16's three-arm first run on SQuAD with intervals. Chapter 17 step 1 done: the gap measured on three tasks and one chosen, or none found.

To 2027-10. Chapters 16 and 17 answered on one gap task: either the loop closes (state return beats text return, the handoff rate falls after learning) or the kill is recorded and the thesis narrows to instrument + trigger + audit. Paper 2, the audit and the return path, submitted. The priced sender catalogue has at least the four existing pairs as rows. `goal/GOAL.md`'s 12-month line is rewritten from "state channels that carry uncertainty" to "populations that learn from returns" only if Chapter 16 survives; otherwise it stays. Prediction 11 scored on its date, 2027-12-31.

To 2029-10. Thesis defended on three papers: the instrument and the message-versus-reader audit (done); the return-path learning rule; the closed loop priced with its fraud detector. One cross-family pair (Qwen3-4B → 7B already works for content) inside the loop, one drift test across a receiver update, Chapter 15 either closed or killed with numbers. Every chapter's controls rerunnable from `analysis/cloud/*.json` and the code already in the repository. Predictions 8 and 3 scored on their dates, 2028-12-31 and 2030-12-31.

18.2 What kills the whole program

Four things, any one of which shrinks the thesis and two of which end the population claim.

1. *Bytes.* The message is 72 MiB per 512-token prefix; layers cannot be cut (18–35 with filler: 28.7%, 16.1 F1), low rank keeps the binding test but not passages (rank 16: 29.6 F1), and int4 reaches only 36 KB per token. [SHOWN] If nothing near 1 MB keeps content, the channel stays an instrument and the "economy" is text plus the probe.
2. *An instruction-tuned receiver that reads "confidence: low" as well as the cache.* The paper's own named first experiment. Then uncertainty does not cross, it is recomputed, and Chapter 14 collapses into Chapter 13.
3. *Returned state teaching the sender no more than returned text.* Chapter 16's kill. Then "learning from what comes back" is distillation, and the title loses "improve."
4. *No task at this scale with a sender–receiver gap worth a handoff.* SQuAD's 3.5 F1 gap already killed milestone 4 once. [SHOWN]

Behind all four sits the futures document’s own thesis: that latent state stays inside a shared-weights boundary for legal, not technical, reasons. If the predictions hold, the population studied here is one owner’s models: a defensible thesis, and not the agent economy of the north star.

18.3 Why it lasts

The question outlives any model pair. Whether a population of models can improve by trading state is not settled by a better projector, and the field’s own audits show that controls, not channels, are the scarce thing: [Cheng et al. \[2026\]](#) find most latent gain on natural benchmarks is receiver computation (MedQA: +0.4 of +14.7 content-specific), and [Zhang and Emu \[2026\]](#)’s boundary-replacement controls are what a reviewer now asks for. A COLM 2026 reviewer of StateBridge wrote that “the paper does not demonstrate latent communication between different models” (`docs/research-program-2026-09-14.md`), and that is the bar every chapter here is built to clear. The repository owns the instrument (binding test, deranged cache, deranged-sender projector, filler fills, within-item intervention) and every later chapter reuses it unchanged, so each result is cheap and comparable to the last. It fits one A100 and a few hundred dollars a month because every experiment is eval-heavy on frozen models with small adapters, as the first 23 blocks were. And the futures document’s fifteen dated predictions give the program external checks that mature on their own calendar, so it is checked against a calendar it does not control.

18.4 What it costs

The repository’s own costed runs are the only price data, and they are quotes, not list prices: `docs/design-cascade-2026-09-24.md` budgets one sender pass and three receiver passes on one A100 at under \$10, and `docs/design-confidence-2026-09-22.md` budgets the confidence protocol at under \$15. The standing GCP authorisation in `goal/GOAL.md` is \$500 total. **[BET]** No hour count appears in any of those files, so none is stated here; the plan per quarter is written in the two quoted units. The under-\$10 quote covers one sender pass and three receiver passes (30 min + 3 × 40 min); the under-\$15 quote covers the full confidence protocol. Chapter 14 is three runs of the under-\$15 quote (one per experiment (i)–(iii)) plus one rerun; Chapter 15 and the drift test of Chapter 17 are each about one under-\$10 run on cached states; Chapters 12 and 13 need no new GPU beyond re-evaluation on cached states and the per-submission receiver pass of the catalogue, whose engineering hours per pair are listed next to its GPU cost when a pair is admitted. Chapter 16’s nine adapter trainings (three arms × three seeds) and Chapter 17’s task runs have no quote in the design documents and are priced before they run. Arithmetic on the quotes: \$500 buys about 33 runs at under \$15 or 50 at under \$10, so the allowance covers the eval-only blocks above and not the unpriced training blocks; each of those is quoted to Peter first, as `goal/GOAL.md` requires. No cloud job was launched for this document.

19. Conclusion

Score the message, then score the reader.

What crossed. Content: 63.0% on an 8-name binding test against 12.5% chance, 53.0 SQuAD F1 against 54.7 for text, and a receiver whose confidence moves within an item because of the cache’s content, Spearman .280 against .004 for another item’s cache. The reading that survives is that evidence crosses and the receiver recomputes.

What did not. The word: a label that scores .955 on its own, read as .552. The bytes: 72 MiB per 512-token message, and no cut below a quarter of that keeps the content. The pay-off: on SQuAD no trigger makes a handoff worth it, because the receiver reading the passage is 55.7 F1 against the sender’s 52.2.

What an economy of state would need. A price sheet, declared here before the run that could change it. A fraud, measured: the receiver follows the wrong episode’s cache 63.7% of the time. An audit, which is this document’s one reusable instrument: score the message, then the reader. A reputation, untested, because no sender in the repository changes.

What would overturn the main claim is a receiver whose entropy follows “confidence: low”: an instruction-tuned or few-shot reader. It is the first experiment the program runs. What would make the population claim true is a sender that learns more from returned state than from returned text; that experiment has not been started, and its kill is written.

The request to the field is one line: any paper that reports a verbal-confidence baseline should report the label-only AUROC next to the receiver AUROC.

Appendices

A. Methods: projector, ridge, training

Models. Sender K2-Horizon-3.7B, receiver K2-Horizon-7B, both frozen, bf16, RoPE $\theta = 10^7$; they share a tokenizer and KV geometry (36 layers, 8 KV heads, head dimension 128). The sender’s pre-RoPE K and raw V are captured at every layer, mapped, RoPE is applied at the receiver, and the result is inserted as the receiver’s cache for the prefix positions. The receiver then reads only the question. For Qwen3-4B as sender (matched geometry, different vocabulary) receiver positions are aligned to sender tokens by character offsets.

Stage 1: ridge. For each receiver layer ℓ and KV head h , a ridge regression from the concatenated $[K; V]$ of the sender (256-d per head) to the receiver’s $[K; V]$. Source layers are the three sender layers whose K/V correlate best with receiver layer ℓ (“top-3”), which beat the aligned-layer map in a first pass (32.7% against 71.3% once trained, Table A.1). Fit on FineWeb-Edu sequences (`k2cascade.projector.ridge, --map topk --k 3 --seqs 512`).

Stage 2: residual. A gated SwiGLU residual (bottleneck 128, 151 M parameters) on top of the ridge output, trained with AdamW for 1,500 steps, effective batch 16 sequences of 1,024 FineWeb-Edu tokens, prefix 512, on the receiver’s cross-entropy over the continuation (`k2cascade.projector.train`). Nothing in training mentions the tests. The ridge is not frozen during training (freezing it: 59.0%). Bottleneck 32 has 70 M parameters (64.3%, 53.7 F1); bottleneck 16 has 62 M (61.3%, 51.6 F1).

Deranged-sender projector. The identical recipe with the sender reading text unrelated to the receiver’s continuation (`--derange_source`); it can learn a soft prompt but not a content channel.

Probes. Failure probe (Table 10.2): logistic regression on the sender’s last-prompt-token hidden state at layers 12 / 18 / 24, five-fold, $C = 0.1$, label “sender F1 < 0.5”. Agent-step probe (`nested_3.7b.json`): nested leave-one-run-out over 26 runs, layer fixed a priori, run-level cluster bootstrap. Four-point uncertainty probes (Table 9.1): standardised logistic probe, $C = 0.05$, five passage-grouped folds, no inner selection, label “sender SE above median”.

Table A.1: Projector ablations on the 6-name Noma variant. All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, $n = 300$ episodes, chance 12.5%. Accuracy in percent. The two 1,000-step rows with an added uncertainty term use the full 151 M-parameter projector.

projector	project	derange	follow
ridge top-3 (closed form)	22.0	10.7	30.3
+ residual, 300 steps	56.3	6.3	53.7
+ residual, 1500 steps, seed 0	71.3	4.3	71.7
+ residual, 1500 steps, seed 1	73.0	3.0	74.0
+ residual, 1500 steps, seed 2	75.3	2.3	76.0
+ residual, 1500 steps, ridge frozen	59.0	7.0	59.3
+ residual, 1500 steps, aligned-layer ridge	32.7	8.0	34.7
+ residual, 1000 steps, bottleneck 32	64.3	5.7	62.3
+ residual, 1000 steps, bottleneck 16	61.3	7.7	60.0
+ residual, 1000 steps, + KL(text cache)	66.0	5.3	63.3
+ residual, 1000 steps, + sender-entropy matching	72.3	3.0	73.3

B. Protocol definitions: arms, variants, metrics

Binding test (Noma). Each episode assigns one of k single-token values to k nonce names; 8-name ($n = 1000$) and 6-name ($n = 300$) variants, chance 12.5% in both. Arms: *none* (question only); *text* (receiver reads the facts); *self* (receiver’s own cache re-injected; equals *text* to 10^{-4}); *raw* (sender cache injected without a map); *project* (mapped cache); *derange* (mapped cache from another episode with the same layout and a different value for the queried name). Content transfer = project – derange. Follow rate: how often, under *derange*, the answer is the other episode’s value.

Passages. SQuAD v1.1 validation; sender reads the passage (≤ 512 tokens), receiver reads only the question, answers greedily (≤ 8 tokens, stopped at newline); token-F1 against gold spans; mean gold log-probability per token. HotpotQA bridge questions ($n = 400$): sender and receiver each hold one supporting paragraph; the receiver always reads its own paragraph and the question.

Uncertainty protocol. Three passage variants per item: clean; contradicted (one fluent counter-answer sentence written by the 375B judge, kept if DeBERTa-MNLI says it entails the counter-answer and not the gold; 587 of 941 survive); removed (gold sentence deleted). Items kept: the 377 (344 passages) on which the receiver reading the clean passage reaches $F1 \geq .5$. Sender target: semantic entropy over 10 samples, thresholded at its median per variant. Receiver signal: first-token entropy. Four messages: text; mapped cache; verbal (greedy answer + “confidence: high / medium / low” from the sender’s actual SE terciles over the 587 items); numeric (sample agreement); candidate list (every cluster with its frequency). Receiver prompt: “A smaller model that read the passage answered: X (confidence: *high*).” followed by the question; no chat template, no few-shot, no hedge instruction. Controls per variant: deranged cache, zero cache, moment-matched random cache, question only.

Metrics. Message-level signal: AUROC of a fixed function of the message alone (word rank; number; for the cache, a linear probe). Receiver-level signal: AUROC of the receiver’s first-token entropy after reading the message. Read-out gap: their difference, comparable within an arm only. Drop ratio: the handoff’s decrease in $P(\text{gold})$ from clean to removed divided by the text path’s decrease. Within-item: per item, Spearman between the receiver’s entropy change and the sender’s SE change when only the sender’s passage changes. Belief shape: per item with at least two sampled clusters, Spearman between

the sender’s cluster frequencies and the receiver’s first-token mass on those candidates. Intervals: paired bootstrap over passages (344 passages, 2,000 resamples) on three cache differences. Retention: $(\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on fresh held-out sequences.

Cost. Bytes per prefix token (KB = 1024 bytes); latency on one A100 for a 512-token prefix; network time at a stated link rate from the message size.

C. Review history: six external reviews and what changed

Six external reviews arrived on 2026-09-25 (`docs/review-external-robotics-2026-09-25.md`); the paper went from v3 to v5 in response. The entries below follow that file’s headings, so “the fourth review” in Chapter 10 and “the same-source baseline” in Chapter 15 point at the entries of the same name.

First review: robotics direction. A structured-text baseline (candidate list with probabilities) was a real hole; added as the candidate-list arm (.495 / .489 / .626). Interlat and StateBridge added to the comparison set. Bytes over a network, not only GPU time, added to the cost paragraph. The next paper’s setting reframed as act / observe-more / ask / stop in a partially observable task, with two kinds of “don’t know.”

Second and third reviews, same day: corrections before arXiv. “All content-free controls sit at .46–.54” was false on the removed variant and is now reported per variant. The theory box’s step from log 3 nats to a lower AUROC was invalid and was withdrawn. The confidence word is an oracle label from the sender’s actual SE terciles, stated as such. The drop ratio is a relative drop in $P(\text{gold})$, not a fraction of uncertainty retained. Heo et al.’s retention is an accuracy ratio, ours a loss-gap ratio, not conflated. The deranged-sender projector is weakly content-dependent (21.7% ridge base), not content-free. K2-0.9B “fails on this pair,” not “requires head dimension 128.” The current setting is context reuse, not a mid-task handoff.

Fourth review: formulas and code. Zero-fill dilutes the softmax, so the filler null (receiver’s own cache of a neutral text) was added and the collapse survived it. Four-point uncertainty probes implemented with passage-grouped folds. Attention-output alignment and cross-head maps noted for the next paper.

Fifth review: audits before anything else. The label-only AUROC computed (`audit_confidence.py`; .955 / .942 / 1.000). Paired passage-grouped bootstrap intervals added; SE discretisation reported (38 / 40 / 28 distinct values; 22% at threshold on removed). The receiver’s text path is an empirical reference, not a ceiling. “Any learned map compresses uncertainty” softened to an observation. Entropy matching is not belief matching; the right target is a separate reader $r(m, q)$. Three goals separated: content reuse, belief recovery, decision quality. Same-source baseline named for the next paper.

Sixth review: FOREWARN to UPS chain. The within-item intervention run the same day (cache .280 / 88%; text .322 / 90%; word .146 / 49%; deranged .004 / 47%). The decision-loss bound ($2R_{\text{max}}\epsilon$) adopted as the modest replacement for the deleted deduction.

Not an external review: a later verifier pass on v5 (commit 3762708) corrected the probe localisation sentence, added the same-sign caveat, fixed Related Work and cut the abstract to 209 words.

What the reviews left standing: the binding test with its derange arm, the deranged-sender projector, the SQuAD and HotpotQA numbers, the cost sweep, and the receiver-level ordering. What they rewrote: every mechanism claim.

D. Reproducibility: commands, runs, costs

Code. Package `k2cascade/projector/`: `ridge.py` (stage 1), `train.py` (stage 2), `noma.py` (binding test), `qa.py` (passages, arms, variants), `eval.py` (retention), `sender_entropy.py`, `verbal.py`, `confidence.py`, `audit_confidence.py`, `diag_uncertainty.py`, `nli_filter.py`, `latency.py`, `cascade.py`, `policy.py`. Agent-step probes: `k2cascade/nested.py`, `shortcut.py`. Job scripts under `cloud/jobs/`: `noma_ridge.sh`, `train_residual.sh`, `program.sh`, `design2.sh`, `design3.sh`, `followup.sh`, `confidence.sh`, `verbal.sh`, `night_a.sh`, `night_b.sh`, `compress.sh`, `xfamily.sh`, `improve.sh`, `cascade.sh`.

Provenance by table. Table 4.1: `analysis/cloud/noma_main_ridge_top3.json`, `noma_main_mlp_top3_s1500_seed0.json`, `noma_main_ctrl_derange_s1500.json`. Seeds and ablations (Table A.1): `prog_noma*.json`, `improve/`. Table 4.2: `noma_layers*.json`. Table 4.3: `ret_fresh*.json`. Table 6.1: `qa_none.json`, `qa_ridge_top3.json`, `qa_mlp_seed2.json`, `qa_mlp_seed2_n600.json`, `qa_ctrl_derange.json`. Table 5.1: `noma_city*.json`, `noma_animal*.json`, `fu_noma_rev*.json`. Table 5.2: `xfamily/`, `night_b/`. Table 6.2: `compress*/`, `night_b/`. Table 5.3: `noma_pad*.json`. Tables 7.1 and 7.2: `compress/`, `compress2/`, `compress3/`. Table 8.1: `conf/`, `verbal/confidence_verbal_mlp_seed2.json`, `night_a/confidence_numeric.json`, `confidence_candidates.json`, `confidence_belief.json`; label-only audit, intervals and within-item: `verbal/audit.json`. Table 10.1: `verbal/confidence_verbal_mlp_kl1.json`, `night_a/confidence_ent1.json`. Table 9.1: `night_a/diag_unc_clean.json`, `diag_unc_removed.json`. Table 10.2: `cascade/curves.json`. Agent-step probe: `nested_3.7b.json`, `shortcut_k2-3.7b.json`. A run-level index is in `analysis/cloud/summary.md`.

Run ids named in the repository. `program-20260922b` (seeds, ablations), `design2-20260922-022708` (controls, attributes, distance, layer bands), `k2-confidence-20260923-084423` (uncertainty protocol), `compress-20260925-173145` (filler-null pruning). Checkpoints: `runs/mlp_top3_s1500_seed0/step_1500` (and seeds 1, 2), `runs/ctrl_derange_s1500/step_1500`, `runs/ridge_top3`.

Costs. All runs on one A100 at a time on GCP. The two design documents are the only costed quotes: the uncertainty protocol at under \$15 (`docs/design-confidence-2026-09-22.md`) and the handoff-policy run at under \$10 (`docs/design-cascade-2026-09-24.md`). The standing allowance is \$500 total (`goal/GOAL.md`). No cloud job was launched for this document; every number in it was read from the files above.

Build. `cd docs/thesis && PATH=/Library/TeX/texbin:$PATH latexmk -pdf thesis.tex`, then `latexmk -c`. Figures are read from `docs/paper/figs/` through the document’s graphics path; Figure 2.1 is drawn in the document from Table 8.1.

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